

Empirical averaging in deterministic optimal control

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Abstract

A windowed averaged scheme is defined for general control systems. The same method is used to average costs in optimal control problems (OCPs). A numerical parameter α can be computed, which expresses the distance between the original system and the averaged system in a weak sense.

Then, if we use the optimal control of the averaged OCP in the original OCP, the suboptimality of the control is bounded by an expression of the form $C\alpha^2$.

1 Introduction

We consider here optimal control problems which feature fast dependency on time. This is the case, for instance, if the problems depend on data which features high frequency phenomenons, i.e., features that happen at a much smaller time scale than the response time of the dynamics.

We consider here the solution of these control problems using the Pontryagin's minimum principle. In this perspective, we have to solve a two point boundary value problem, be it through direct or indirect methods. To do that, we have to discretize the boundary problem. In our case, the size of the mesh is dictated by the high frequency features of the data, which is much smaller than the time scale of the dynamics. The question that is addressed in this paper is to find a solution method whose mesh size is dictated by the dynamics and not by the data.

When the fast data is periodic with respect to time, a classic approach to the solving of differential equations is averaging. Historically, the method of averaging was introduced to study the motion of celestial bodies by solving a simple two body equation which is perturbed by the influence of other bodies (see the section in [28] about the history of averaging). As developed in [25], the framework was that of the perturbation of an orbit by the small influence of a periodic input. Averaging was then generalized in a geometric framework,

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notably in [1, 2]. A comprehensive book on the subject is [28]. In this framework, the faster oscillating phenomenon is either periodic or satisfies some kind of ergodicity assumption.

The previous references consider a non controlled dynamical systems. We turn our attention now to controlled systems, and, in particular, optimal control problems (OCPs).

In this perspective, we add to the differential equation an extra parameter u which is the open loop control of the system. If we have non-linearities in the dynamics, common sense dictates that averaging be performed *after* the control is introduced, by contrast to the approach where the control would be introduced *after* the differential system is averaged. Indeed, the high frequency content of the control may interact with the high frequency content of the data; therefore the averaging must be applied once this interaction is taken into account.

In the case of ordinary differential equations, the approximation value of the averaging method is determined by the proximity of the nominal and averaged trajectories.

However, in the case of controlled systems, there are as many trajectories to consider as there are different controls. Fortunately, in optimal control, there is a simple way to evaluate the performance of an approximation method:

- We first consider the optimal cost J_0 for the nominal control system.
- On the other hand, we considered the optimal control u^* for the approximated control system, and we use it to control the *nominal* system. This generates a cost J^* .
- Naturally, J^* is greater than J_0 . As a consequence, we can evaluate the quality of the approximation method by considering the loss of optimality $J^* - J_0$ which is non negative.

Let us review some previous work on this topic.

An early work [3] applies averaging to two point boundary value problems, but its application to optimal control is limited to what is essentially the linear quadratic case. In [10], the method of averaging is applied to optimal control, both in open loop (the system is then periodic with respect to the fast time), and in closed loop (which is a study of the Hamilton-Jacobi equation under an ergodicity assumption). The study of the HJB equation is improved in [4]. Observe that, in these two references, the horizon is finite and the oscillatory input is fast.

References [17, 16, 14, 15] consider an optimal control problem “in the long run”, that is, with slow and “normal” time scales, using averaging techniques. The convergence of the optimal cost is proved, but there is no estimation of the loss of optimality that the approximation method produces.

Reference [23] constructs the method of averaging for controlled systems on time scales, and determines the algorithm of correspondence of controls over the original and averaged systems. It is an interesting work, but we shall prove in this article that the order of efficiency of the method is under-estimated (it should be two).

As we pointed out before, the application of averaging to celestial mechanics is a classic. It has been naturally transposed to the context of optimal control. Specifically, the optimal control of low thrust engines in space have been studied

from a geometric view point and in the periodic case. Such a approach has been adopted in [7, 6, 8]. Averaging has also been used for similar problems [18, 13] in a spirit that is close to [10]. All of these references assume that the fast dynamics are either periodic or satisfy an ergodicity assumption.

There are applications of averaging in optimal control that are more data driven (by contrast to autonomous celestial mechanics). Energy management of thermal systems consider the optimal control of the temperature of buildings [27]. Naturally, this heavily depends of the weather data, which naturally features fast oscillating components (with respect to the building's inertia).

Reference [12] considers a buoy which is subject to the oscillating influence of waves.

Averaging has been also used in stochastic optimal control, notably of Markov chains (see for instance [20, 33, 31]). Indeed, when there exists a cycle in a discrete state Markov chain that has high transition probabilities, then this cycle is gone through very fast and averaging can be applied. An alternative to the classical averaging scheme is ergodic theory [26].

In this article, we consider optimal control problems which are influenced by some fast perturbation data (with respect to the time constant of the system). The approach is somehow different from what exists in the literature.

Here we consider, as far as rapidly oscillating inputs are considered, that the behavior of the system is similar to the response of a single integrator. This is why we first define our averaging method by the effect of integration on general signals. The averaged signal is defined by its successive averages on consecutive intervals. This transformation can be from various points of view:

- In the frequency domain, this averaging method is low pass filter, although not a very good one [30, 29, 24]. Indeed, it is a Finite Impulse Response filter, and thus its transfer function is not compactly supported.
- In the time domain, this averaging operator is an orthogonal projection on the Haar basis. More efficient approximation methods exist in the wavelet framework [22]; however specific tools must be used for compactly supported signals [11, 32] and, in practice, their have boundary effects that are not desirable when noise is present.
- From a statistical point of view, this method can be viewed as the computation of a sequence of empirical expectations, typically for a non stationary signal. This is why we call our method *empirical* averaging.

From this approximation technique we derive a similar one for ordinary differential equations, control systems, and ultimately optimal control problems.

In all of these contexts, the performance of the averaging method is evaluated by the difference between the output of the reference signal and the averaged one, through an integrator. The maximum value for this difference is called α .

It is a very simple thing to evaluate and as a consequence, it may be applied to a wide range of situations.

An important feature of this problem is that the mesh size of the two point boundary value problem is not dictated by the fast data, but rather by the time constant of the system (which is larger than the width of the averaging interval). The minimization consists in the minimization of an averaged Hamiltonian.

This numerical technique has already been applied to the optimal control of a low thrust space engine that must reach its final orbit; the controls in the

Keplerian case (where periodicity exists) and in the non Keplerian case (where the influence of the moon and of the sun breaks the periodicity) have been successfully computed [19].

The paper is organized as follows. Section 2 presents the original OCP. It then presents windowed averaging for functions, differential equations and control systems. It then presents the concept of averaged OCP that is used in this paper. This is where the averaging error α is defined. Section 3 describes the example of the orbit transfer of low thrust space engine in the non keplerian case [19]. Sections 4 and C compute formal expansions of the state and of the costate of the nominal problem with respect to the parameter α . Auxilliary variables that will be used later in the course of the article are defined here. Section 5 introduces an auxiliary problem of optimization as well as new auxiliary variables. The main assumptions (bounded derivatives and convexity) are given before we state the auxiliary problem. The main result is exposed in section 6. It is a result on the control cost, trajectories and the optimal control. Section 7 is devoted to the proof of the main theorem. A conclusion is presented in section 8.

Note that, for a fluent reading of the paper, complicated computations are detailed in the appendices.

2 Problem statement

2.1 Nominal Problem

We wish to solve the following Optimal Control Problem (OCP) :

$$\min_u \int_0^T L(x, u, t) dt \quad (1)$$

where x is a finite dimensional state which satisfies the dynamics :

$$\frac{dx}{dt} = f(x, u, t), \quad x(0) = \chi_0 \quad (2)$$

where u is an integrable finite dimensional and unconstrained control. The assumptions on f and L are given in section 5.1.

2.2 Averaged Problem

If the dependency of f or L with respect to time contains fast oscillations, the nominal problem may be difficult to solve because the mesh size is dictated by the fast oscillation rate. To avoid this, we define an averaged problem with the help of a very simple low pass filter. This averaging method applies to any kind of time-varying (or data-driven) system.

2.2.1 Filtering of a signal by windowed averaging

We consider a signal defined on an interval and split this interval into N consecutive intervals. On each interval we compute the average of the signal.

Definition 1. Let N an integer, and g an integrable function on $[0, T]$. Let us subdivide $[0, T]$ into N intervals $[t_k, t_{k+1}]$, with:

$$t_k = k \frac{T}{N}, \quad k = 0..N \quad (3)$$

The averages of g on these intervals define the low pass filter LP on g :

$$LP[g](t) = \frac{1}{t_{k+1} - t_k} \int_{t_k}^{t_{k+1}} g(s) ds, \quad t \in [t_k, t_{k+1}), \quad k = 0..N-1 \quad (4)$$

The difference $g - LP[g]$, that is the averaging error, defines the high pass filter HP on g :

$$HP = Id - LP \quad (5)$$

We then denote $I[g]$ the antiderivative of the averaging error:

$$I[g](t) = \int_0^t HP[g](s) ds, \quad t \in [0, T] \quad (6)$$

We show now that the upper bound of the function $I[g]$ is small when N is big and g is bounded.

Proposition 1. The following bound holds for any bounded function g and any $t \in [0, T]$:

$$|I[g](t)| \leq 2\|g\|_\infty \frac{T}{N} \quad (7)$$

Proof. Let us first prove that for any $k = 0..N$

$$\int_{t_k}^{t_{k+1}} HP[g](t) dt = 0$$

Indeed:

$$\int_{t_k}^{t_{k+1}} HP[g](t) dt = \int_{t_k}^{t_{k+1}} g(t) dt - \int_{t_k}^{t_{k+1}} \left[\frac{1}{t_{k+1} - t_k} \int_{t_k}^{t_{k+1}} g(s) ds \right] dt = 0$$

Hence, for $t \in [t_k, t_{k+1}[$:

$$I[g](t) = \int_{t_k}^t HP[g](s) ds$$

But $|LP[g](s)| \leq \|g\|_\infty$, so that $|HP[g](s)| \leq 2\|g\|_\infty$. This gives finally:

$$|I[g](t)| \leq 2\|g\|_\infty (t - t_{k+1}) \leq 2\|g\|_\infty \frac{T}{N}$$

□

Thus we may define a small number α that represents how close the signal g is closed to its averaged signal $LP[g]$:

$$\alpha = \|I[g]\|_\infty \quad (8)$$

Example 1 Let us suppose that g is periodical with period $\frac{T}{N}$. Then $LP[g]$ is constant equal to the mean of g over a period. The function $HP[g]$ is a periodic signal with 0 average. The function $I[g]$ is the periodic antiderivative of $HP(g)$ with value 0 at 0. Its upper bound α is of order $\frac{T}{N}$.

Example 2 Let us suppose that g is periodical with period ϵ a small divisor of T , and $N = 1$. Then $LP[g]$ is constant equal to the mean of g over the many periods in $[0, T]$. The function $HP[g]$ is a periodic signal of small period ϵ with 0 average. The function $I[g]$ is the periodic antiderivative of $HP(g)$ with value 0 at 0. Its upper bound α is of order ϵ . This example shows that α can be small with seemingly large intervals.

2.2.2 Averaging errors for ordinary differential equation

Let us consider the ordinary differential equation (ODE):

$$\frac{dx}{dt} = f(x, t), \quad x(0) = \chi_0 \quad (9)$$

with f Lipschitz with respect to x and integrable with respect to t .

The averaged ODE is defined as:

$$\frac{dx_0}{dt} = LP[f](x_0, t), \quad x_0(0) = \chi_0 \quad (10)$$

where the low-pass filter LP on a general function $g(x, t)$ is defined as:

$$LP[g](\xi, t) = \frac{1}{t_{k+1} - t_k} \int_{t_k}^{t_{k+1}} g(\xi, s) ds, \quad t \in [t_k, t_{k+1}), \quad k = 0..N-1 \quad (11)$$

As a summary, $LP[g]$ averages g with respect to time on rectangular adjacent windows, leaving the state variable unchanged.

The state x_0 is well defined and bounded because f , and thus $LP[f]$, is Lipschitz.

We consider now x_0 the solution of (10) and $g(x, t)$ a general function of the state and time. We establish a bound in the following proposition that will be used for the definition of the small parameter α , which will measure in section 2.2.6 the averaging error for optimal control problems.

Proposition 2. *Let $g(x, t)$ be bounded Lipschitz in x with a Lipschitz constant λ_g , and let's suppose that $f(x, t)$ is bounded. Let $I[g, x_0]$ the function of t defined by:*

$$I[g, x_0](t) = \int_0^t [g(x_0(s), s) - LP[g](x_0(s), s)] ds \quad (12)$$

where x_0 is the solution of the averaged ODE (10) Then the following bound holds:

$$|I[g](t)| \leq 2(\|g\|_\infty + \lambda_g T \|f\|_\infty) \frac{T}{N} \quad (13)$$

Proof. see appendix A. □

We now estimate the distance between the trajectories obtained respectively from the nominal and averaged system.

Proposition 3. *Let x defined by the ODE (9) and let x_0 defined by the averaged ODE (10). Let λ the Lipschitz constant of f in x and define*

$$\alpha = \sup_{t \in [0, T]} (|I[f](t)|)$$

Then the following bound holds:

$$\|x - x_0\|_\infty \leq \alpha \frac{e^{\lambda T} - 1}{\lambda} \quad (14)$$

In other words, the averaging error $\|x - x_0\|_\infty$ is in first order in α , that is small when N is big because f is Lipschitz and thus bounded on $[0, T]$.

Proof. We have

$$\begin{aligned} |x(t) - x_0(t)| &= \left| \int_0^t f(x(s), s) - f(x_0(s), s) + f(x_0(s), s) - LP[f](x_0(s)) ds \right| \\ &\leq \lambda \int_0^t |x(s) - x_0(s)| ds + \alpha \end{aligned}$$

which leads to the result thanks to the Gronwall lemma. $\square$

2.2.3 Averaging of a controlled system

In the scope of this article, we define, for the integer $N > 0$, and any functions $u(t)$ and $g(x, u(t), t)$, an averaged function $LP[g, u]$

$$\begin{aligned} LP[g, u](x, t) &= \frac{N}{T} \int_{\frac{kT}{N}}^{\frac{(k+1)T}{N}} g(x, u(s), s) ds \\ &\text{for } t \in \left[\frac{kT}{N}, \frac{(k+1)T}{N} \right) \text{ and } k \in [0, N-1] \end{aligned} \quad (15)$$

As a summary, $LP[g]$ averages g with respect to time (this includes the open loop control) on rectangular adjacent windows, leaving the state (or costate) unchanged.

2.2.4 Statement of the averaged problem

We define the averaged OCP which consists in minimizing the cost

$$J_0(v) = \int_0^T LP[L, v](y(s), s) ds \quad (16)$$

where y is the state defined by

$$\frac{dy}{dt} = LP[f, v](y, t), \quad y(0) = \xi_0 \quad (17)$$

The system (17) is a well defined differential equation.

Assumption 1. *The averaged OCP admits an optimal control u_0 with a corresponding trajectory x_0 .*

The optimal trajectory x_0 is defined by the ODE:

$$\frac{dx_0}{dt} = LP[f, u_0](x_0, t), \quad x_0(0) = \xi_0 \quad (18)$$

2.2.5 Stationnarity condition for the averaged problem

Theorem 1. *Let H be the Hamiltonian of the nominal problem:*

$$H(x, u, p, t) = L(x, u, t) + p f(x, u, t) \quad (19)$$

Let p_0 be the costate of the averaged problem, defined along the optimal trajectory x_0 by the backwards differential equation:

$$\frac{dp_0}{dt} = -LP \left[\frac{\partial H}{\partial x}, u_0 \right] (x_0, p_0, t), \quad p_0(T) = 0 \quad (20)$$

Then the following stationarity condition holds:

$$\frac{\partial H}{\partial u}(x_0(t), u_0(t), p_0(t), t) = 0 \text{ a.e.} \quad (21)$$

where a.e. stands for almost everywhere in $t \in [0, T]$.

Proof. see appendix B □

2.2.6 Introduction of a small α

We want to define a small α as we have done for signals and differential equations.

For a function $g(x, u, t)$, the difference $g(x_0(t), u_0(t), t) - LP[g, u_0](x_0(t), t)$ is the result of high pass filtering $HP[g, u](x_0(t), t)$. Let's define the antiderivative:

$$I[g, x_0, u_0](t) = \int_0^t HP[g, u_0](x_0(s), s) ds \quad (22)$$

Then, as a consequence of Proposition 2, the following bound holds:

$$|I[g, x_0, u_0](t)| \leq (2\|g\|_\infty + \lambda_g T \|f\|_\infty) \frac{T}{N} \quad (23)$$

In other words, $\|I[g, x_0, u_0]\|_\infty$ can be made small if N is large, f is bounded and g is bounded and Lipschitz.

Let's define similarly the backwards antiderivative

$$I^T[g, x_0, u_0](t) = \int_t^T HP[g, u_0](x_0(s), s) ds \quad (24)$$

Then, with a similar proof, the same inequality holds:

$$|I^T[g, x_0, u_0](t)| \leq (2\|g\|_\infty + \lambda_g T \|f\|_\infty) \frac{T}{N} \quad (25)$$

Consequently, $\|I^T[g, x_0, u_0]\|_\infty$ can also be made small if N is large, f is bounded and g is bounded and Lipschitz.

Definition 2. *For the rest of that document, we consider the small number α defined as:*

$$\alpha = \sup \left(\|I[f, x_0, u_0]\|_\infty, \left\| I^T \left[\frac{\partial H}{\partial x}, (x_0, p_0), u_0 \right] \right\|_\infty \right) \quad (26)$$

We shall see that α measures how close the nominal and averaged problems are. Observe that α is small, for instance, when N is big and assumption 2 below holds (bounded functions and their derivatives).

Observe that that α depends only on the solution of the averaged problem.

2.3 Numerical perspective

Technically, the proof of our main approximation result is based on the Pontryagin principle. This is a theoretical result which holds independently of the actual numerical method used to solve the nominal (resp. averaged) problem.

Now that we have stated what the averaged optimal control problem is, we can begin to evaluate its benefit when solving numerically a general optimal control problem.

Let us consider a direct method for the solving of problem (1,2). To do so, we discretize the dynamics into

$$x_{k+1} = x_k + \delta t f(x_k, u_k, t_k) \quad (27)$$

In order to represent accurately the influence of the data on the system, the discretization step δt must be a suitable sampling rate for the data. In this article's framework, δt is much smaller than the time constant of the system. The approximation of the cost is obtained by using a quadrature rule for the integral. Without using extra knowledge on the system, the time step of the quadrature is δt .

In this framework, the optimal control problem becomes a finite dimensional optimization problem with a finite number of equality constraints.

We consider now a similar direct method to solve the averaged problem (16,17). To avoid confusion, we denote by capital T_K the boundaries of the averaging intervals, and X_K denotes the state of the discretized averaged system. The dynamics of X is

$$X_{K+1} = X_K + \int_{T_K}^{T_{K+1}} f(X_K, u(s), s) ds \quad (28)$$

where a suitable quadrature rule is used to compute the integral. Consistently with the discretization (27), we consider the sampling rate of the control and data to be δt . The computation of the cost is handled similarly.

We see that, while (27) only consider the differential aspect of the dynamics, the averaged dynamics (28) incorporates the integral action of a dynamical system.

Let us now consider the benefit of solving the averaged problem versus solving the nominal problem. To make things even simpler, let us use a Riemann integral for the computation of (28) and define

$$R = \frac{T_{K+1} - T_K}{\delta t}$$

We assume that R is a (large) integer. Indeed, δt is dictated by the sampling rate of the data, while $T_{K+1} - T_K$ is dictated by the time constant of the system, which is much larger.

We see that the number of control variables u_k is the same in both problems. However, the *number of constraints* (28) for the averaged problem is R times smaller than the number of constraints (27) for the original problem. This is the main benefit of using averaging to solve the optimal control problem with direct methods.

Finally, observe that the definition of the averaged costate (20) indicates what should be an indirect method, i.e. based on the Pontryagin principle, for the solving of the averaged problem. In this case, the benefit of using the averaging method is a coarser grid for the two point boundary value problem.

3 Applicative Example

We describe here the example of the orbit transfer of low thrust space engine in the non keplerian case [19]. Ionic thrust engines work as follows: instead of using a chemical reaction to eject material, ions are ejected from the engines using an eletro-magnetic field. The resulting thrust is much lower than with a clas-sical engine, but the required mass for the same acceleration is proportionnally smaller (within the operating range).

3.1 Description of the problem

A geostationary satellite (communication, meteo, TV) can not reach its orbit directly with a classical launcher. It is rather set on a transfer elliptic orbit and then brought to its geostationnary orbit by its own means: see figure 1.

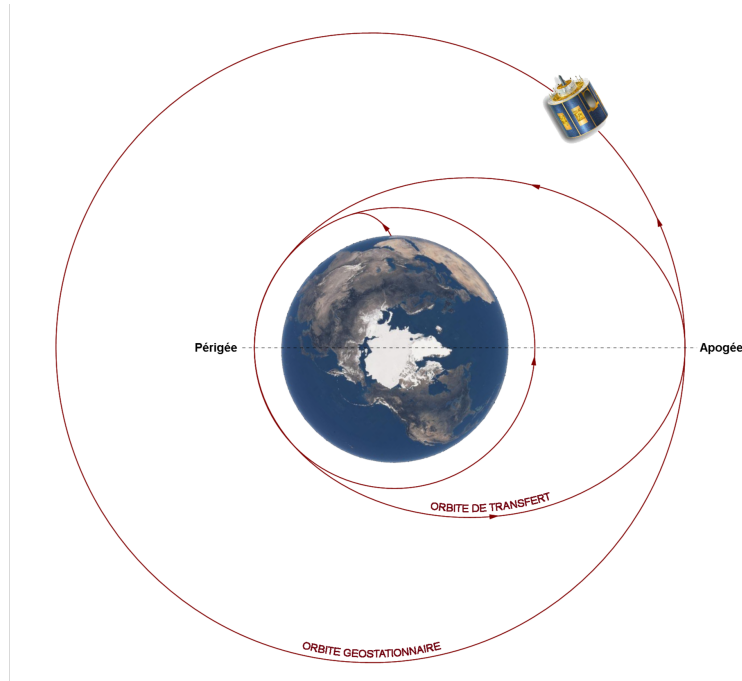


Figure 1: Setting a geostationnary satellite to its operational orbit

The orbit transfer is done by a ionic thrust engine rather than a conventional engine in order to save the mass of fuel and oxygen needed by a conventional engine. Thus the transfer is slow and needs a lot of orbits (about 90).

The optimisation criteria is either the time length of the orbit transfer or the mass loss and thus the initial mass need.

3.2 Characteristics of the optimal control problem

The dynamics of the satellite is very close to an orbital movement, so that the parameters may be divided into:

- orbital slow parameters,
- angular speed position on the orbit,
- plus the mass, that is also slowly decreasing.

Thus the problem is ill conditioned. Indeed, the optimal control problem, with a 90 orbits horizon, needs a very great amount of time samples, so that the two boundary value problem can not be reasonably numerically solved.

3.3 Averaging on successive orbits

In order to take advantage of the almost orbital movement, an averaging process is done on each orbit, so that the time sampling can be coarser and the optimal control problem can be numerically solved.

Note that a time conversion is done, defining the angular position on the orbit as the new “time”, so that the dynamic is nearly periodic of period 2π .

3.3.1 Keplerian Case

When only the earth gravity is taken into account, we are in the so called Keplerian case, and the dynamic is periodic, with the low thrust of the ionic engine, the control parameter, as only perturbation. In that case, a “classical” averaging scheme on successive orbits can be performed.

Result The solution of the averaged problem is a progressive modification of the orbit until it is the geostationary one, as shown on Figure 2.

The costs of the orbital transfer are respectively a duration of 1235.4457 days and a mass loss of 208.7603 kilos.

The thrust strategy is divided into two phases, with a transition at about half the orbital transfer:

- an acceleration phase, where the satellite is accelerated at its apogee and perigee, so that apogee and perigee heights are both increased,
- a deceleration phase, where the satellite is decelerated at its apogee and accelerated at its perigee, so that apogee and perigee heights are respectively decreased and increased.

3.3.2 Non-Keplerian Case

Moon and sun gravity forces can not be neglected in the context of a low thrust transfer. So that they are taken into account in a further version of the optimal orbit transfer search.

But this results in a non-Keplerian problem, that is only nearly periodic and not exactly periodic in the angular position on the orbit.

This is why a windowed averaging scheme on 2π wide successive windows is interesting. But we can wonder whether that averaging scheme leads to costs and thrust strategy that are close to the Keplerian case.

The answer is *yes*:

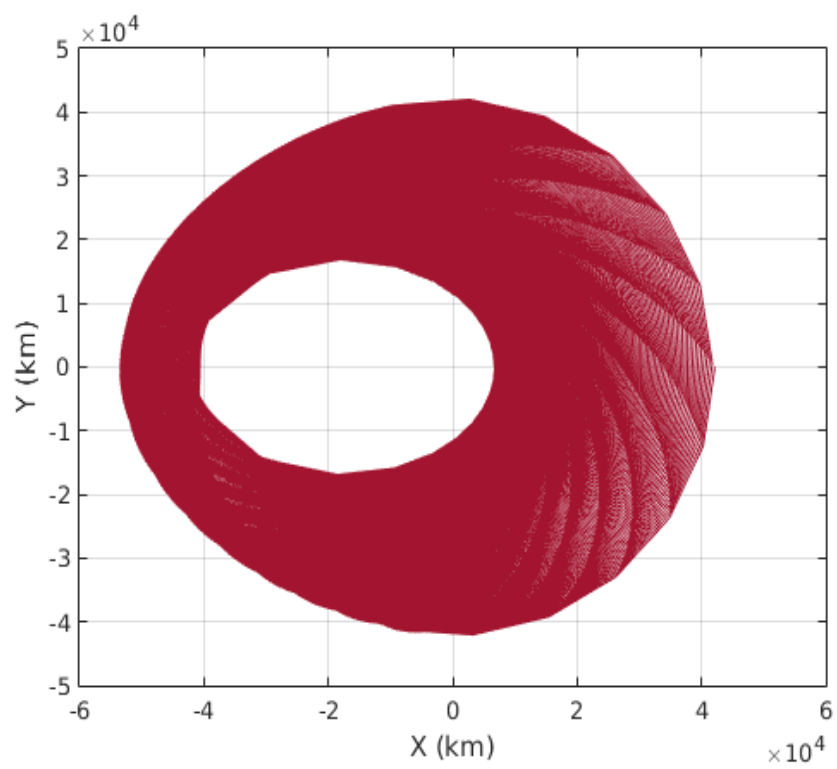


Figure 2: Successive orbits during the orbital transfer in the Keplerian Case

- the costs of the optimal averaged problem of orbital transfer are respectively a duration of 135.0423 days and a mass loss of 208.1386 kilos, that are very close to the corresponding costs in the Keplerian case.
- The thrust strategy is divided into the same two phases, acceleration and then deceleration, as in the Keplerian phase

This is true even though the costates are quite different in the different configurations.

3.4 Conclusion of the example

The widowed averaging scheme proves to be as affective in the periodic case (Keplerian) than in the non-periodic case (non-Keplerian case with moon and sun gravity forces). And this is very useful in order to have an accurate *and* numerically solvable optimal orbit transfer problem. The fact that the solution is nearly optimal, with close to each other trajectories, is the result of the present paper.

4 A priori expansions and definitions of auxiliary variables

We are now going to define auxiliary variables by means of formal expansions with respect to α . In what follows, u_0 , x_0 and p_0 denote the optimal control, state and costate of the averaged problem as described in section 2.2.4.

4.1 Notations

Definition 3. We denote the following variables from the state x (resp. x_0), the control u (resp. u_0) and the costate p (p_0):

$$\begin{aligned}\sigma &= (x, u) , \quad \sigma_0 = (x_0, u_0) \\ w &= (x, u, p) , \quad w_0 = (x_0, u_0, p_0)\end{aligned}$$

where p is the costate of the nominal problem defined by the ODE with final condition:

$$\frac{dp}{dt} = -\frac{\partial H}{\partial x}(x, u, p, t) , \quad p(T) = 0 \quad (29)$$

Definition 4. We denote the derivatives up to second order of functions with respect to the variables x , u or σ with indexes, on the model:

$$\begin{aligned}f_x &= \frac{\partial f}{\partial x} \\ H_{uu} &= \frac{\partial^2 f}{\partial u^2} \\ H_{\sigma\sigma} &= \begin{bmatrix} H_{xx} & H_{xu} \\ H_{ux} & H_{uu} \end{bmatrix}\end{aligned}$$

4.2 Formal expansion in α

The variable α defined by the equation (26) is viewed as a small quantity¹. In Appendix C we compute a formal expansion of the state x and the costate p at the first order with respect to α .

These formal expansions lead to the definition of auxiliary variables $\tilde{x}_1$ and $\tilde{p}_1$ as follows:

Definition 5. *We define the auxiliary variables $\tilde{x}_1$ and $\tilde{p}_1$ the following way:*

$$\alpha \tilde{x}_1 = I[f, x_0, u_0] \quad (30)$$

and

$$\alpha \tilde{p}_1 = I^T[H_x, (x_0, p_0), u_0]$$

The error terms $\tilde{x}_1$ and $\tilde{p}_1$ correspond to the deviation error due to the approximation of the dynamics with respect to their fast oscillating behavior.

In definition 8 of section 5.3 we define deviations that are defined in *magnitude*, by contrast to the sole averaging error (at the optimum) which is considered here.

A consequence of the definition of $\alpha \tilde{x}_1$ in definition 5 and α in definition 2 is that:

$$\|\tilde{x}_1\|_\infty \leq 1 \quad (31)$$

And a consequence of the definition of $\alpha \tilde{p}_1$ in definition(5) and α in definition 2 is that:

$$\|\tilde{p}_1\|_\infty \leq 1 \quad (32)$$

5 Auxiliary Problem

We define an auxiliary problem which is similar to the study of the second variation in [9].

5.1 Assumptions

We make two kinds of assumption on the problem data; the first is related to the regularity of the functions involved, and the second is a convexity assumption (Legendre-Clebsch) that is common in optimal control [9, 5, 10].

Assumption 2 (Regularity). *The derivatives, up to the third order, of f and L with respect to x and u are bounded by some $k > 0$.*

A consequence of that assumption is that the value α defined by equation (26) is well defined and small if N is sufficiently big (or if the problem is periodic of small period).

As a consequence, we can bound the second derivative of the dynamics f :

Proposition 4.

$$\left| f_{\sigma\sigma}(X, U, t) \begin{bmatrix} Y \\ V \end{bmatrix}^2 \right| \leq k(|Y| + |V|)^2 \quad (33)$$

for any (X, U, Y, V) .

¹Actually it is not required that α be small. Its purpose is to give a one dimensional direction to a variation in a high dimensional space.

Proof.

$$\begin{aligned} \left| f_{\sigma\sigma}(X, U, t) \begin{bmatrix} Y \\ V \end{bmatrix}^2 \right| &= |f_{xx}(X, U, t)Y^2 + 2f_{xu}(X, U, t)YV + f_{uu}(X, U, t)V^2| \\ &\leq k(|Y| + |V|)^2 \end{aligned}$$

□

Moreover, as p_0 is the solution of the ODE with final condition (20), it is differentiable and thus continuous of the bounded interval $[0, T]$, so that it is bounded. So that another consequence of the assumption 2 involves the derivatives of the Hamiltonian:

Proposition 5. *The hamiltonian $H(x, u, p_0, t)$ and its derivatives up to the third order in u and x are bounded by a constant K .*

Proof. Take $K = (1 + \|p_0\|_\infty)k$.

□

As a consequence, the Hessian of the Hamiltonian is bounded:

Proposition 6.

$$\left| H_{\sigma\sigma}(X, U, p_0 t) \begin{bmatrix} Y \\ V \end{bmatrix}^2 \right| \leq K(|Y| + |V|)^2 \quad (34)$$

for any (X, U, Y, V) .

Proof. Similar as for $f_{\sigma\sigma}$.

□

Assumption 3 (Strong Legendre-Clebsch condition [9]). *There exists $\beta > 0$ so that for any (x, u) , the following holds:*

$$H_{uu}(x, u, p_0, t) \geq \beta Id \quad (35)$$

and

$$[H_{xx} - H_{xu}(H_{uu})^{-1}H_{ux}](x, u, p_0, t) \geq 0 \quad (36)$$

The consequence of equation (35) is that H_{uu} is invertible and $\|H_{uu}^{-1}\|_\infty \leq \frac{1}{\beta}$ on any (x, u, p_0) .

The consequence of equation (36) is :

Proposition 7.

$$H_{\sigma\sigma}(x, u, p_0, t) \geq 0 \quad (37)$$

Proof. see Appendix D

□

5.2 Auxiliary Problem Definition and Properties

Definition 6. *Let's define the following notations:*

$$H_{0\sigma\sigma} = H_{\sigma\sigma}(w_0, t) = \begin{bmatrix} H_{0xx} & H_{0xu} \\ H_{0ux} & H_{0uu} \end{bmatrix}$$

and

$$f_{0x} = f_x(\sigma_0, t), \quad f_{0u} = f_u(\sigma_0, t)$$

Then we define the auxiliary problem as the linear quadratic OCP for the state y steered by the control v :

$$\frac{dy}{dt} = f_{0x}[\tilde{x}_1 + y] + f_{0u}v, \quad y(0) = 0 \quad (38)$$

$$J_1(v) = \int_0^T \left[\frac{1}{2} \begin{bmatrix} y & v \end{bmatrix} H_{0\sigma\sigma} \begin{bmatrix} y \\ v \end{bmatrix} + \tilde{p}_1 (f_{0x}y + f_{0u}v) \right] dt \quad (39)$$

where $\tilde{x}_1$ and $\tilde{p}_1$ are defined in definition 5 as the averaging errors on the dynamics of the optimal state and costate at the optimal solution of the averaged problem (as defined in section 2.2.4).

Proposition 8. *There exists an optimal cost v_1 for the auxiliary problem.*

Proof. It is a convex linear quadratic problem because $H_{0\sigma\sigma}$ is non-negative (equation (37)). $\square$

Definition 7. *We denote v_1 an optimal control of the auxiliary problem, y_1 the trajectory corresponding to v_1 and q_1 the corresponding costate.*

Proposition 9. *y_1 follows the dynamics:*

$$\frac{dy_1}{dt} = f_{0x}(y_1 + \tilde{x}_1) + f_{0u}v_1, \quad y_1(0) = 0 \quad (40)$$

Proof. This is the definition of the dynamics of the auxiliary problem. $\square$

Moreover, we have:

Proposition 10. *The stationarity condition $\frac{\partial H_1}{\partial u}(y_1, v_1, p_1) = 0$ may be written the following way:*

$$H_{0xu}y_1 + H_{0uu}v_1 + (\tilde{p}_1 + q_1)f_{0u} = 0 \quad (41)$$

Moreover, the costate q_1 of the auxiliary problem follows the dynamics:

$$\frac{dq_1}{dt} = -H_{0xx}y_1 - H_{0xu}v_1 - (\tilde{p}_1 + q_1)f_{0x}, \quad q_1(T) = 0 \quad (42)$$

Proof. The hamiltonian of the auxiliary problem expands in:

$$\begin{aligned} H_1(y, v, q) = & \frac{1}{2} (H_{0xx}y^2 + 2H_{0xu}yv + H_{0uu}v^2) \\ & + \tilde{p}_1 (f_{0x}y + f_{0u}v) + q_1 [f_{0x}(y + \tilde{x}_1) + f_{0u}v] \end{aligned} \quad (43)$$

$\square$

Finally, we have:

Proposition 11. *y_1, v_1 and q_1 are bounded by a constant M .*

Proof. The auxiliary problem is smooth and convex. $\square$

5.3 More auxiliary variables and their upper bounds

Definition 8. For any $u \in \mathbf{L}_{[0,T]}^2$, x is the trajectory of the nominal dynamics (2).

Let's then define the following notations:

$$\begin{aligned}\delta x &= x - x_0 & \tilde{x} &= \delta x - \alpha \tilde{x}_1 \\ \delta u &= u - u_0 & \tilde{u} &= \delta u \\ \delta \sigma &= (\delta x, \delta u) & \rho(\lambda, \mu) &= \sigma_0 + \lambda \mu \delta \sigma\end{aligned}$$

Where (u_0, x_0) is a solution of the averaged problem and $\alpha \tilde{x}_1$ is $I[f, x_0, u_0]$ (equation (30)).

The δ variables are deviations in magnitude around the optimal control u_0 , after a correction due to the pure averaging error on the various dynamics.

Definition 9. We define the following variables:

$$\begin{aligned}r &= \tilde{x} - \alpha y_1, \quad v = \tilde{u} - \alpha v_1 \\ Z[\lambda, \mu](t) &= v + [H_{uu}^{-1} H_{ux}](\rho(\lambda, \mu), p_0, t)(r + \alpha \tilde{x}_1) \\ z^2 &= \int_0^1 \int_0^1 \lambda \|Z[\lambda, \mu]\|_2^2 d\lambda d\mu\end{aligned}$$

Here r and v are variations with respect to the solution of the averaged optimal control problem, with the extra corrective terms

- the averaging error on the optimal dynamics *considered as a pure signal*
- the correction term that corresponds to the second variation in reference [9], as defined in definition 6.

The quantity z^2 is repeatedly used in [5] to study perturbations in optimal control; it has been used later in [21] in a similar manner.

We will bound $\|r\|_\infty^2$ and $\|v\|_2^2$ with respect to z^2 and α^2 . Bounding then z^2 with respect to α^2 will lead to the main result.

Definition 10. We define the following constants:

$$\begin{aligned}k_{r1} &= 4k^2 T e^{2k(1+\frac{K}{\beta})} \\ k_{r2} &= \frac{\left(\frac{K}{\beta} + \frac{1}{2}(1+2M)^2\alpha\right)^2}{\left(1 + \frac{K}{\beta}\right)^2} \left(e^{k(1+\frac{K}{\beta})} - 1\right)^2 \\ k_{v1} &= 6 \left(2 + \frac{K^2 T k_{r1}}{\beta^2}\right) \\ k_{v2} &= \frac{6K^2 T (k_{r2} + 1)}{\beta^2}\end{aligned}$$

with:

- k is introduced in assumption 2 about the boundedness of the derivatives of $f(x, u, t)$ and $L(x, u, t)$.

- $K = (1 + \|p_0\|_\infty)k$
- M is the upper bound of the optimal trajectory of the auxiliary problem introduced in section 5.2 (proposition 11).
- α is the (hopefully) small quantity defined in equation (26).
- β is the convexity constant of H_{uu} introduced in equation (35) inside of the Legendre-Clebsch conditions of assumption 3.

Proposition 12. *The following inequalities hold:*

$$\|r\|_\infty^2 \leq k_{r1}z^2 + k_{r2}\alpha^2 \quad (44)$$

and:

$$\|v\|_2^2 \leq k_{v1}z^2 + k_{v2}\alpha^2 \quad (45)$$

Proof. see appendix E.1 □

We now define an extra correction term that will be used further on. We will show that this correction term is also bounded by values that depend only on z and α .

Definition 11. r_1 is defined by the following dynamics:

$$\begin{aligned} \frac{dr_1}{dt} &= f(r_1 + x_0 + \alpha(\tilde{x}_1 + y_1), v + u_0 + \alpha v_1, t) \\ &\quad - f(x_0 + \alpha(\tilde{x}_1 + y_1), u_0 + \alpha v_1, t) \\ r_1(0) &= 0 \end{aligned} \quad (46)$$

The state r_1 is driven by the difference between the dynamics with state r_1 and input v , and the nominal velocity of the system for the value of the state and controls that include all the previously introduced correction terms.

We will bound $\|r - r_1\|_\infty$ in α^2 and $\|r_1\|_\infty^2$ in z^2 and α^2 . As a consequence, we will “anchor” our error estimates to the nominal term r_1 .

Definition 12. *We define the following constants:*

$$k_{r5} = \frac{T}{2}(1 + 2M)^2(e^{kT} - 1), \quad k_{r3} = 2k_{r1}, \quad k_{r4} = 2(k_{r2} + k_{r5}^2\alpha^2)$$

with k , K and M as in definition 10.

Proposition 13. *The following inequalities hold:*

$$\|r - r_1\|_\infty \leq k_{r5}\alpha^2 \quad (47)$$

and

$$\|r_1\|_\infty^2 \leq k_{r3}z^2 + k_{r4}\alpha^2 \quad (48)$$

Proof. see appendix E.2 □

6 Main theorem

Definition 13. We define the following constants:

$$\begin{aligned}
k_{J1} &= 4\sqrt{3}KMTk_{r1} + \left(\frac{3k}{2} + KM\right)Tk_{r3} + \left(\frac{3k}{2} + (2\sqrt{3} + 1)KM\right)k_{v1} \\
k_{J2} &= 2\sqrt{3}KTk_{r5} + k(1 + 2M) + 2KTM(M + 1) \\
&\quad + \frac{3k}{2}Tk_{r2} + \left(kM + 4\sqrt{3}KM\right)Tk_{r4} + \left(\frac{3k}{2} + (2\sqrt{3} + 1)KM\right)k_{v2} \\
k_{J0} &= \left[e^{kT} - 1 + \frac{K}{2k^2T}(e^{kT} - 1)^2 + \frac{k}{2}\alpha\right] \\
k_J &= k_{J2} + k_{J0} \\
k_x &= \sqrt{2\left(\frac{2k_Jk_{r1}}{\beta} + k_{r2} + (M + 1)^2\right)} \\
k_u &= \sqrt{2\left(\frac{2k_Jk_{v1}}{\beta} + k_{v2} + T^2M^2\right)}
\end{aligned}$$

where:

- k, K, M, α and β are as in definition 10.
- k_{r1}, k_{r2}, k_{v1} and k_{r2} are defined in definition 10
- k_{r3}, k_{r4} and k_{r5} are defined in definition 12.

Note that these constants depend only of k, K, M, α, β , and the horizon T .

Assumption 4.

$$\alpha \leq \frac{\beta}{2k_{J1}}$$

Theorem 2 (Main Theorem). *Considering the nominal problem in section 2.1, let $H(x, u, p, t)$ its Hamiltonian, and let $J^* = \inf_u J(u)$ be its infimum cost.*

Let u_0 a solution of the averaged problem described in section 2.2.4 with its trajectory x_0 . Such a solution exists by assumption 1.

Let α be the small quantity defined in equation (26) and let β be the constant introduced in assumption 3.

Let the set of constants (k_{J1}, k_J, k_x, k_u) introduced in definition 13.

Then, under the set of assumptions listed in section 5.1 and the assumption 4, the following inequalities hold:

- the suboptimality of the real system commanded by u_0 is limited to:

$$J^* \leq J(u_0) \leq J^* + k_J\alpha^2 \quad (49)$$

- any trajectory x of the nominal problem for a u better than u_0 ($J(u) \leq J(u_0)$), is close to (x_0, u_0) , with:

$$\|x - x_0\|_\infty \leq k_x\alpha \quad (50)$$

$$\|u - u_0\|_2 \leq k_u\alpha \quad (51)$$

7 Proof of the main result

7.1 Proof Process

To prove the main theorem, we proceed the following way.

The section 7.2 is devoted to the search of a lower bound of any real cost $J(u)$ of the nominal problem. That lower bound contains two integral terms that do not depend on u , a term in z^2 that is the only one depending on u , and a term in α^2 , that does not depend on u either.

The section 7.3 is devoted to the search of an upper bound of the real cost $J(u_0)$ of the nominal problem controlled by u_0 . That upper bound contains the same two integral terms as in the lower bound of $J(u)$ and a term in α^2 .

Then the section 7.4.1 uses the assumption $\alpha \leq \frac{\beta}{k_{J1}}$ to obtain a lower bound of $J(u)$ independent of u , so that it is also a lower bound for J^* . That lower bound is combined with the upper bound of $J(u_0)$ to bound the suboptimality $J(u_0) - J^*$ as a function of α^2 , as stated in the equation (49) of the first part of the main theorem 2.

Then the section 7.4.1 uses the stronger assumption $\alpha \leq \frac{\beta}{2k_{J1}}$ to obtain, for any u better than u_0 , i.e. so that $J(u) \leq J(u_0)$, an upper bound of z^2 . With that bound of z^2 , upper bounds for $\|x - x_0\|_\infty$ and $\|u - u_0\|_2$ are found in the equations (50) and (51) of the second part of the main theorem, with the help of the definitions and bounds of r and v in section 5.3.

7.2 Lower bound on the real cost of the nominal problem

7.2.1 Expansion of the real cost $J(u)$

We use an expansion with integral remainder to compute the variation of the cost with respect to the nominal integral cost for the nominal system being driven by u_0 .

Proposition 14. *The cost $J(u) = \int_0^T L(x, u, t)dt$ for any command $u \in \mathbf{L}_{[0,T]}^2$ expands the following way:*

$$\begin{aligned} J(u) = & \int_0^T L(x_0, u_0, t)dt + \alpha \int_0^T H_{0x} \tilde{x}_1 dt + \int_0^T HP[H_x, u_0](x_0, p_0, t) \tilde{x} dt \\ & + \int_0^T \int_0^1 \int_0^1 \lambda H_{\sigma\sigma}(\rho(\lambda, \mu), p_0, t) (\delta\sigma)^2 d\lambda d\mu dt \end{aligned} \quad (52)$$

Proof. see appendix F. □

We are going now to study carefully the various terms involved in the expansion (52).

7.2.2 Lower Bound of the third term of the expansion of $J(u)$ in equation (52)

Proposition 15. *The following inequality holds:*

$$\begin{aligned}
& \int_0^T HP[H_x, u_0](x_0, p_0, t) \tilde{x} dt \\
& \geq \alpha \int_0^T \tilde{p}_1 [f_{0x} r_1 + f_{0u} v] dt - \frac{3k}{2} (Tk_{r3} + k_{v1}) \alpha z^2 \\
& - \left[2KTk_{r5} + k(1 + 2M) + \frac{k}{2} (3(Tk_{r4} + k_{v2}) + 4T(1 + 2M)^2) \alpha \right] \alpha^2
\end{aligned} \tag{53}$$

Proof. See appendix G. $\square$

7.2.3 Lower Bound of the fourth term of the expansion of $J(u)$ in equation (52)

Proposition 16. *The following inequality holds:*

$$\begin{aligned}
& \int_0^T \int_0^1 \int_0^1 \lambda H_{\sigma\sigma}(\rho(\lambda, \mu), p_0, t) (\delta\sigma)^2 d\lambda d\mu dt \\
& \geq -2KM \left[TM + \sqrt{3}(2Tk_{r2} + k_{v2} + T(M^2 + 2)) \alpha \right] \alpha^2 \\
& + \left[\beta - 2\sqrt{3}KM(2Tk_{r1} + k_{v1}) \alpha \right] z^2 \\
& + \alpha \int_0^T \tilde{p}_1 [y_1, v_1] H_{0\sigma\sigma} \left[\begin{array}{c} r + \alpha \tilde{x}_1 \\ v \end{array} \right] dt
\end{aligned} \tag{54}$$

Proof. see appendix H. $\square$

7.2.4 Bound in absolute value for the sum of the integral terms of the right hand sides of equations (53) and (54)

Proposition 17. *Let's define R as the sum of the integral terms of the right hand sides of equations (53) and (54):*

$$R = \alpha \int_0^T \tilde{p}_1 \left[f_{0x} r_1 + f_{0u} v + [y_1, v_1] H_{0\sigma\sigma} \left[\begin{array}{c} r + \alpha \tilde{x}_1 \\ v \end{array} \right] \right] dt$$

Then the following inequality holds:

$$|R| \leq KM[Tk_{r3} + k_{v1}] + [2KTM + (2KTMk_{r5} + kTMk_{r4} + kMk_{v2}) \alpha] \alpha^2 \tag{55}$$

Proof. see appendix I. $\square$

7.2.5 Lower Bound of the real cost $J(u)$

We capitalize on the previous results to obtain a lower bound on $J(u)$ for any control u .

Lemma 1. *A lower bound of the cost $J(u)$ of the nominal system for any control u is given by:*

$$J(u) \geq \int_0^T L(x_0, u_0, t) dt + \alpha \int_0^T H_{0x} \tilde{x}_1 dt + (\beta - k_{J1} \alpha) z^2 - k_{J2} \alpha^2 \quad (56)$$

where k_{J1} and k_{J2} are defined in definition 13.

Proof. This is a consequence of equations (53), (54) and (55) $\square$

7.3 Upper bound on the cost of the nominal system controlled by u_0

As in the previous section, we first compute an expansion with integral remainder of the cost; then we study each term of the expansion to obtain a global bound in section 7.3.5.

7.3.1 Expansion of the cost $J(u_0)$

Definition 14. *Let x^0 be the trajectory of the nominal problem controlled by u_0 . It is defined by the dynamics:*

$$\frac{dx^0}{dt} = f(x^0, u_0, t), \quad x^0(0) = \chi_0 \quad (57)$$

For that trajectory, we set the notations:

$$\begin{aligned} \delta x^0 &= x^0 - x_0 & \tilde{x}^0 &= \delta x^0 - \alpha \tilde{x}_1 \\ \rho^0(\lambda, \mu) &= x_0 + \lambda \mu \delta x^0 \end{aligned}$$

Where (u_0, x_0) is the solution of the averaged problem and $\alpha \tilde{x}_1$ is $I[f, x_0, u_0]$ (equation (30)).

Proposition 18. *The cost $J(u_0) = \int_0^T L(x^0, u_0, t) dt$ for the optimal command u_0 of the averaged problem expands the following way:*

$$\begin{aligned} J(u_0) &= \int_0^T L(x_0, u_0, t) dt + \alpha \int_0^T H_{0x} \tilde{x}_1 dt + \int_0^T HP[H_x, u_0](x_0, p_0, t) \tilde{x}^0 dt \\ &\quad + \int_0^T \int_0^1 \int_0^1 \lambda H_{xx}(\rho^0(\lambda, \mu), u_0, p_0, t) (\delta x^0)^2 d\lambda d\mu dt \end{aligned} \quad (58)$$

Proof. It is a consequence of proposition 14 with $u = u_0$, so that $\delta u = 0$. $\square$

7.3.2 Comparison of x^0 and x_0

Proposition 19. *The following inequality hold for $\delta x^0 = x^0 - x_0$:*

$$|\delta x^0| \leq \frac{\alpha}{kT} (e^{kT} - 1) \quad (59)$$

Proof. As $\delta x^0 = x^0 - x_0$, x^0 follows the dynamics (57) and x_0 follows the averaged dynamics (18), we have the following integral equation:

$$\begin{aligned}\delta x^0 &= \int_0^t [f(x^0, u_0, t) - LP[f, u_0](x_0, t)] dt \\ &= \int_0^t [f(x^0, u_0, t) - f(x_0, u_0, t)] dt + \int_0^t [f(x_0, u_0, t) - LP[f, u_0](x_0, t)] dt\end{aligned}$$

Thus the following inequality holds:

$$|\delta x^0| \leq \|f_x\|_\infty \left(\int_0^t |\delta x^0| ds \right) + |I[f, x_0, u_0](t)| \leq kT \left(\int_0^t |\delta x^0| ds \right) + \alpha$$

Equation (59) follows from Gronwall lemma. $\square$

7.3.3 Upper Bound of the third term of the development of $J(u_0)$ (58)

Proposition 20. *The following inequality holds:*

$$\left| \int_0^T HP[H_x, u_0](x_0, p_0, t) \tilde{x}^0 dt \right| \leq \left(e^{kT} - 1 + \frac{k}{2} \alpha \right) \alpha^2 \quad (60)$$

Proof. By definition of $\tilde{x}^0$, we have

$$\tilde{x}^0 = x^0 - x_0 - \alpha \tilde{x}_1,$$

so that:

$$\begin{aligned}\frac{d\tilde{x}^0}{dt} &= f(x_0 + \delta x^0, u_0, t) - LP[f, u_0](x_0, t) + HP[f, u_0](x_0, t) \\ &= f(x_0 + \delta x^0, u_0, t) - f(x_0, u_0, t)\end{aligned}$$

Thus, by Taylor expansion of $f(x^0, u_0, t)$ with integral remainder, we have:

$$\frac{d\tilde{x}^0}{dt} = f_x(x_0, u_0, t) \delta x^0 + \int_0^1 \int_0^1 \lambda f_{xx}((x_0 + \lambda \mu(\delta x^0), u_0, t) (\delta x^0)^2 d\lambda d\mu$$

Thus, thanks to proposition 19, we have:

$$\left| \frac{d\tilde{x}^0}{dt} \right| \leq (e^{kT} - 1) \alpha + \frac{k}{2} \alpha^2 \quad (61)$$

But an integration by part, together with the fact that $HP[H_x, u_0](x_0, p_0, t) = -\alpha \frac{d\tilde{p}_1}{dt}$ and that $\tilde{x}^0(0) = \tilde{p}_1(T) = 0$ leads to:

$$\int_0^T HP[H_x, u_0](x_0, p_0, t) \tilde{x}^0 dt = \alpha \int_0^T \tilde{p}_1 \frac{d\tilde{x}^0}{dt} dt \quad (62)$$

Including equation (61) and the fact that $\|\tilde{p}_1\|_\infty \leq 1$ into equation (61) proves equation (60) $\square$

7.3.4 Upper Bound of the fourth term of the development of $J(u_0)$ (58)

Proposition 21. *The following inequality holds:*

$$\left| \int_0^T \int_0^1 \int_0^1 \lambda H_{xx}(\rho^0(\lambda, \mu), u_0, p_0, t) (\delta x^0)^2 d\lambda d\mu dt \right| \leq \frac{K}{2k^2T} (e^{kT} - 1)^2 \alpha^2 \quad (63)$$

Proof. This is a consequence of the proposition 19. $\square$

7.3.5 Upper Bound of $J(u_0)$

Inserting equations (60) and (63) into equation (58) leads to:

Lemma 2. *An upper bound of the cost $J(u_0)$ of the nominal system controlled by u_0 is given by:*

$$J(u_0) \leq \int_0^T L(x_0, u_0, t) dt + \alpha \int_0^T H_{0x} \tilde{x}_1 dt + k_{J0} \alpha^2 \quad (64)$$

where k_{J0} is defined in definition 13.

7.4 Proof of the main theorem

The main theorem is proved in two steps. The first step compares the costs to estimate the suboptimality of the real system controlled by u_0 . The second step compares the trajectories and controls which outperform u_0 , to estimate how close they are from the trajectory and control driven by u_0 .

7.4.1 Comparison of the real cost controlled by u_0 and the infimum cost of the real system

Let's now use the assumption $\alpha \leq \frac{\beta}{2k_{J1}}$ of the first part of the Main Theorem 2 into the equation (56) of Lemma 1. The term in z^2 is then non negative, and we get the lower bound independent of u :

$$J(u) \geq \int_0^T L(x_0, u_0, t) dt + \alpha \int_0^T H_{0x} \tilde{x}_1 dt - k_{J2} \alpha^2$$

As that lower bound holds for any u , it is also a lower bound for the infimum cost $J^* = \inf_u J(u)$:

$$J^* \geq \int_0^T L(x_0, u_0, t) dt + \alpha \int_0^T H_{0x} \tilde{x}_1 dt - k_{J2} \alpha^2$$

so that:

$$\int_0^T L(x_0, u_0, t) dt + \alpha \int_0^T H_{0x} \tilde{x}_1 dt \leq J^* + k_{J2} \alpha^2$$

Inserting this equation into the equation (64) of Lemma 2, together with the fact that $J(u_0) \geq J^*$, by definition of J^* , proves the suboptimality equation (49) in the Main Theorem 2, since $k_J = k_{J0} + k_{J2}$.

7.4.2 Comparison of the controls and trajectories with and without $u = u_0$

Let's consider a control u better than u_0 , i.e. such that $J(u) \leq J(u_0)$.

Let's now use in a stronger manner the assumption $\alpha \leq \frac{\beta}{2k_{J1}}$ of the second part of the Main Theorem 2 into the equation (56) of Lemma 1. The coefficient z^2 is then lower than $\frac{\beta}{2}$, and we get the lower bound dependent of z^2 , that depends on u :

$$J(u) \geq \int_0^T L(x_0, u_0, t)dt + \alpha \int_0^T H_{0x} \tilde{x}_1 dt + \frac{\beta}{2} z^2 - k_{J2} \alpha^2$$

Thus, together with the equation (64) of Lemma 2, we have the list of inequalities:

$$\begin{aligned} \int_0^T L(x_0, u_0, t)dt + \alpha \int_0^T H_{0x} \tilde{x}_1 dt + \frac{\beta}{2} z^2 - k_{J2} \alpha^2 &\leq J(u) \leq J(u_0) \\ &\leq \int_0^T L(x_0, u_0, t)dt + \alpha \int_0^T H_{0x} \tilde{x}_1 dt + k_{J0} \alpha^2 \end{aligned}$$

So that, with the definition of $k_J = K_{J0} + k_{J2}$, we have

$$z^2 \leq \frac{2k_J}{\beta} \alpha^2 \quad (65)$$

On the other hand, by definition of r , we have:

$$x - x_0 = r + \alpha(\tilde{x}_1 + y_1)$$

so that, together with the equation (44):

$$\|x - x_0\|_\infty^2 \leq 2(\|r\|_\infty^2 + \alpha^2(1 + M)^2) \leq 2(k_{r1} z^2 + (k_{r2}(1 + M)^2) \alpha^2)$$

Introducing equation (65) into that equation leads to equation (50) of the second part of the Main Theorem 2.

Now let's consider the fact that, by definition of v :

$$u - u_0 = v + \alpha v_1$$

so that:

$$\|u - u_0\|_2^2 \leq 2(\|v\|_2^2 + \alpha^2 T^2 M^2) \leq 2(k_{v1} z^2 + (k_{v2} T^2 M)^2 \alpha^2)$$

Introducing equation (65) into that equation leads to equation (51) of the second part of the Main Theorem 2.

8 Conclusion

We have shown that averaging on consecutive intervals is a flexible and efficient way to approximate signals while getting rid of rapidly varying features. We have extended this method to ordinary differential equations, control systems and optimal control problems. This provides a very flexible framework for the approximation of optimal control problems with rapidly varying features.

Indeed, if α is a measure of the averaging error on the dynamics, we have shown that the level of suboptimality due to the use of the solution of the averaged problem into the original control system is bound by some $k\alpha^2$, where k depends on very general bounds on the dynamics and on the convexity of the system.

This makes it possible to use a coarser time mesh in the numerical computation of the solution of the optimal control problem, as explained in section 2.3.

However, this does not decrease the number of optimizations of the Hamiltonian. This stresses, if needed, the requirement for fast minimization routines in this kind of optimal control problem solving.

As a final note, we can observe that the control here is not constrained. However, using an interior penalty approach as in [21] should be a reasonable method to generalize this article's result to the constrained input case.

A Proof of Proposition 2 about the upper bound of $I[g, x_0](t)$

$I[g, x_0](t)$ is defined by equation (12) and x_0 is the solution of the averaged ODE (10). We are going to bound $I[g, x_0]$.

Let's fix t and let K be so that $t \in [t_K, t_{K+1})$. Then we have:

$$I[g, x_0](t) = \sum_{k=0}^{K-1} I_k[g, x_0](t_{k+1}) + I_K[g, x_0](t) \quad (66)$$

where $I_j[g, x_0](\tau)$ is defined for $\tau \in [t_j, t_{j+1})$ as:

$$I_j[g, x_0](\tau) = \int_{t_j}^{\tau} \left[g(x_0(s), s) - \frac{1}{t_{j+1} - t_j} \int_{t_j}^{t_{j+1}} g(x_0(s), \sigma) d\sigma \right] ds$$

We can expand the terms $I_k[g, x_0](t_{k+1})$ and $I_K[g, x_0](t)$ by subtracting and adding the values at t_k .

$$\begin{aligned} I_k[g, x_0](t_{k+1}) &= \int_{t_k}^{t_{k+1}} \left[g(x_0(t_k, s) - \frac{1}{t_{k+1} - t_k} \int_{t_k}^{t_{k+1}} g(x_0(t_k), \sigma) d\sigma \right] ds \\ &\quad + \int_{t_k}^{t_{k+1}} [g(x_0(s), s) - g(x_0(t_k), s)] ds \\ &\quad + \frac{1}{t_{k+1} - t_k} \int_{t_k}^{t_{k+1}} \int_{t_k}^{t_{k+1}} [g(x_0(s), \sigma) - g(x_0(t_k), \sigma)] d\sigma ds \end{aligned}$$

The first term is equal to 0.

The terms integrated once and twice in the second and third terms are both lower or equal to $\lambda_g \|x_0(s) - x_0(t_k)\|$ because g is Lipschitz in x , with Lipschitz constant λ_g .

Moreover, because x_0 is the solution of the averaged ODE (10), we have:

$$\|x_0(s) - x_0(t_k)\| \leq \|f\|_{\infty}(s - t_k)$$

Thus, we have:

$$|I_k[g, x_0](t_{k+1})| \leq 2\lambda_g \|f\|_\infty \int_{t_k}^{t_{k+1}} (s - t_k) ds$$

That is:

$$|I_k[g, x_0](t_{k+1})| \leq 2\lambda_g \|f\|_\infty \left(\frac{T}{N}\right)^2 \quad (67)$$

Let's now expand $I_K[g, x_0](t)$ in an analog way starting from t_K :

$$\begin{aligned} I_K[g, x_0](t) &= \int_{t_K}^t \left[g(x_0(t_K, s)) - \frac{1}{t_{K+1} - t_K} \int_{t_K}^{t_{K+1}} g(x_0(t_K), \sigma) d\sigma \right] ds \\ &\quad + \int_{t_K}^t [g(x_0(s), s) - g(x_0(t_K), s)] ds \\ &\quad + \frac{1}{t_{K+1} - t_K} \int_{t_K}^t \int_{t_K}^{t_{K+1}} [g(x_0(s), \sigma) - g(x_0(t_K), \sigma)] d\sigma ds \end{aligned}$$

The first term is not 0 here, but it is lower or equal to $2\|g\|_\infty \frac{T}{N}$, and the other terms are bounded as the ones of $I_k[g, x_0](t_{k+1})$.

Thus we have:

$$|I_K[g, x_0](t)| \leq \|g\|_\infty \frac{T}{N} + 2\lambda_g \|f\|_\infty \left(\frac{T}{N}\right)^2 \quad (68)$$

Finally, because $K + 1 \leq N$, the inequations (67) and (68) yield to the proposition 2.

B Proof of the stationarity condition of the averaged problem

B.1 Averaged two boundaries problem

Let u_0 be the optimal control of the averaged problem (17) and let x_0 be the corresponding trajectory. x_0 is defined by the ODE with initial condition (18). Let p_0 be the costate of the optimal trajectory, defined by the ODE with final condition (20), with H the hamiltonian (19).

The system constituted of the of equations (18) and (20) is a two boundaries problem. It is defined as the two boundaries problem corresponding to the averaged optimal control problem.

B.2 First variation in the direction of δu

Let $\delta u \in \mathbf{L}_{[0, T]}^2$ a scalar square integrable function on $[0, T]$.

Let $\epsilon > 0$ and let $u_\epsilon = u_0 + \epsilon \delta u$ the variation of u_0 in the direction of δu .

Since $J_0(u_0) = \min_u J_0(u)$, the following stationarity condition holds:

$$\forall \delta u \in \mathbf{L}_{[0, T]}^2, \left(\frac{dJ_0(u_\epsilon)}{d\epsilon} \right)_{\epsilon=0} = 0 \quad (69)$$

Let x_ϵ be the trajectory corresponding to u_ϵ , defined by the dynamics equation 17 with $v = u_\epsilon$:

$$\frac{dx_\epsilon}{dt} = LP[f, u_\epsilon](x_\epsilon, t), \quad x_\epsilon(0) = \xi_0 \quad (70)$$

Let δx be the variation trajectory corresponding to the direction δu given by $\delta x = \left(\frac{dx_\epsilon}{d\epsilon}\right)_{\epsilon=0}$.

Lemma 3. δx respects the following dynamics function:

$$\frac{d(\delta x)}{dt} = LP\left[\frac{\partial f}{\partial x}, u_0\right](x_0, t)\delta x + LP\left[\frac{\partial f}{\partial u}\delta u, u_0\right](x_0, t) \quad (71)$$

Proof.

$$\frac{d(\delta x)}{dt} = \frac{d}{dt}\left(\frac{dx_\epsilon}{d\epsilon}\right)_{\epsilon=0} = \left(\frac{d}{d\epsilon}\frac{dx_\epsilon}{dt}\right)_{\epsilon=0} = \left(\frac{d}{d\epsilon}LP[f, u_0 + \epsilon\delta u](x_\epsilon, t)\right)_{\epsilon=0}$$

Let k be so that $[t_k, t_{k+1})$. Then:

$$\begin{aligned} \frac{d(\delta x)}{dt} &= \left(\frac{d}{d\epsilon}\left(\frac{1}{t_{k+1} - t_k} + \int_{t_k}^{t_{k+1}} f(x_\epsilon(t), u_0(s) + \epsilon\delta u(s), s)ds\right)\right)_{\epsilon=0} \\ &= \frac{1}{t_{k+1} - t_k} + \int_{t_k}^{t_{k+1}} \frac{d}{d\epsilon}(f(x_\epsilon(t), u_0(s) + \epsilon\delta u(s), s))_{\epsilon=0} ds \\ &= \frac{1}{t_{k+1} - t_k} \int_{t_k}^{t_{k+1}} \left(\frac{\partial f}{\partial x}(x_0(t), u_0(s), s)\delta x(t) + \frac{\partial f}{\partial u}(x_0(t), u_0(s), s)\delta u(s)\right) ds \\ &= LP\left[\frac{\partial f}{\partial x}, u_0\right](x_0, t)\delta x + LP\left[\frac{\partial f}{\partial u}\delta u, u_0\right](x_0, t) \end{aligned}$$

□

B.3 Proof of the stationarity result

Let's make use of Equation (69) in developing $\left(\frac{dJ_0(u_\epsilon)}{d\epsilon}\right)_{\epsilon=0}$ for a given δu .

Lemma 4. The derivative at 0 of $J_0(u_\epsilon)$ in ϵ is related to the hamiltonian by the following equation:

$$\left(\frac{dJ_0(u_\epsilon)}{d\epsilon}\right)_{\epsilon=0} = \int_0^T LP\left[\frac{\partial H}{\partial u}\delta u, u_0\right](x_0, p_0, t)dt \quad (72)$$

Proof. Let's use the equation (16) and then commute the differentiation and integration:

$$\left(\frac{dJ_0(u_\epsilon)}{d\epsilon}\right)_{\epsilon=0} = \left(\frac{d}{d\epsilon}\left[\int_0^T LP[L, u_\epsilon](x_\epsilon, t)dt\right]\right)_{\epsilon=0} = \int_0^T \left(\frac{d}{d\epsilon}[LP[L, u_\epsilon](x_\epsilon, t)]\right)_{\epsilon=0} dt$$

For any $k \in [0, N-1]$ and for any $t \in [t_k, t_{k+1})$, we have the definition (15) of LP :

$$LP[L, u_\epsilon](x_\epsilon, t) = \frac{1}{t_{k+1} - t_k} \int_{t_k}^{t_{k+1}} L(x_\epsilon(t), u_\epsilon(s), s)ds$$

Thus if we commute again the integration and the differentiation:

$$\left(\frac{d}{d\epsilon} [LP[L, u_\epsilon](x_\epsilon, t)] \right)_{\epsilon=0} = \frac{1}{t_{k+1} - t_k} \int_{t_k}^{t_{k+1}} \left(\frac{dL(x_\epsilon(t), u_\epsilon(s), s)}{d\epsilon} \right)_{\epsilon=0} ds$$

But by definition of x_ϵ , u_ϵ , δx and δu , we have:

$$\left(\frac{dL(x_\epsilon(t), u_\epsilon(s), s)}{d\epsilon} \right)_{\epsilon=0} = \frac{\partial L}{\partial x}(x_0(t), u_0(s), s) \delta x(t) + \frac{\partial L}{\partial u}(x_0(t), u_0(s), s) \delta u(s)$$

Thus by averaging on $[t_k, t_{k+1}]$, the result is (LP is linear):

$$\left(\frac{d}{d\epsilon} [LP[L, u_0](x_\epsilon, t)] \right)_{\epsilon=0} = LP \left[\frac{\partial L}{\partial x}, u_0 \right] (x_0(t), t) \delta x(t) + LP \left[\frac{\partial L}{\partial u} \delta u, u_0 \right] (x_0(t), t)$$

But because of the dynamics (20) of the averaged costate p_0 , we have, with the definition (19) of the Hamiltonian:

$$LP \left[\frac{\partial L}{\partial x}, u_0 \right] (x_0(t), t) = -\frac{dp_0}{dt} - p_0 LP \left[\frac{\partial f}{\partial x}, u_0 \right] (x_0(t), t)$$

Thus integrating $\left(\frac{d}{d\epsilon} [LP[L, u_0](x_\epsilon, t)] \right)_{\epsilon=0}$ between 0 and T , we obtain:

$$\left(\frac{dJ_0(u_\epsilon)}{d\epsilon} \right)_{\epsilon=0} = \int_0^T -\frac{dp_0}{dt} \delta x - \int_0^T p_0 LP \left[\frac{\partial f}{\partial x}, u_0 \right] (x_0(t), t) \delta x(t) dt + \int_0^T LP \left[\frac{\partial L}{\partial u} \delta u, u_0 \right] (x_0(t), t) dt \quad (73)$$

Let's make an integration by part for the first term of that equation:

$$\int_0^T -\frac{dp_0}{dt} \delta x = -[p_0 \delta x]_0^T + \int_0^T \frac{d(\delta x)}{dt} p_0$$

The variation of $p_0 \delta x$ between 0 and T is null because $\delta x(0) = 0$ and $p_0(T) = 0$. Thus, with the dynamics of δx given by the Lemma 3, the following holds:

$$\int_0^T -\frac{dp_0}{dt} \delta x = \int_0^T p_0 LP \left[\frac{\partial f}{\partial x}, u_0 \right] (x_0, t) \delta x(t) dt + \int_0^T p_0 LP \left[\frac{\partial f}{\partial u} \delta u, u_0 \right] (x_0, t) dt$$

Let's insert this equation in the first term of equation (73). It results in:

$$\left(\frac{dJ_0(u_\epsilon)}{d\epsilon} \right)_{\epsilon=0} = \int_0^T \left[LP \left[\frac{\partial L}{\partial u} \delta u, u_0 \right] (x_0, t) + p_0 LP \left[\frac{\partial f}{\partial u} \delta u, u_0 \right] (x_0, t) \right] dt$$

This proves the equation (72) by definition of the Hamiltonian. $\square$

Let's now make use of equation (72). Let's first fix t and let k be so that $t \in [T_k, t_{k+1})$. Then we have:

$$LP \left[\frac{\partial H}{\partial u} \delta u, u_0 \right] (x_0, p_0, t) = \frac{1}{t_{k+1} - t_k} \int_{t_k}^{t_{k+1}} \frac{\partial H}{\partial u}(x_0(t), u_0(s), p_0(t), t) \delta u(s) ds$$

Let's specialize δu as a "needle variation":

$$\delta u = \frac{t_{k+1} - t_k}{\eta} \mathbf{1}_{[t, t+\eta]} \delta v$$

with $\eta > 0$ so that $t + \eta < t_{k+1}$ and $\delta v \in \mathbf{L}_{[0,T]}^2$.

Then we have:

$$LP \left[\frac{\partial H}{\partial u} \delta u, u_0 \right] (x_0, p_0, t) = \frac{1}{\eta} \int_t^{t+\eta} \frac{\partial H}{\partial u}(x_0(t), u_0(s), p_0(t), t) \delta v(s) ds \xrightarrow[\eta \rightarrow 0]{a.e.} \frac{\partial H}{\partial u}(x_0(t), u_0(t), p_0(t), t) \delta v(t)$$

More precisely, the limit is the value of the function at t everywhere the function is continue, that is for any t possibly except for a countable number of “jumps”. As any countable set is negligible, the limit holds almost everywhere.

Thus, because of the equations (69) and (72), we have:

$$\int_0^T \frac{\partial H}{\partial u}(x_0(t), u_0(t), p_0(t), t) \delta v(t) dt = 0$$

and this is true for any $\delta v \in \mathbf{L}_{[0,T]}^2$. This proves the stationnarity result:

$$\frac{\partial H}{\partial u}(x_0(t), u_0(t), p_0(t), t) = 0 \text{ a.e.}$$

Remark To prove this result, we have used the fact that the control is unconstrained. The extension of the main result to the constrained case can be considered using the same penalty approach as in [21]. In this approach, the constrained problem is approximated by a sequence of unconstrained problems, where the stationarity condition holds.

C Formal expansions in α

We formally expand the state, costate and control variables with respect to α . This step leads to the *definitions* of the various terms which are used throughout the article to finally yield the bounds stated in the main theorem.

C.1 Expansion of the state x

The state variable x is the solution of the original dynamics equation (2) driven by the control u . Let us define a formal expansion of x and u with regards to α :

$$x = x_0 + \alpha x_1 \tag{74}$$

$$u = u_0 + \alpha u_1 \tag{75}$$

Note that the redundant definitions of x_0 and u_0 are consistent, as will be seen later.

We now separate the expansion terms x_0 and x_1 of the state into low (LP) and high (HP) frequencies components:

$$x_0 = \bar{x}_0 + \tilde{x}_0, \quad x_1 = \bar{x}_1 + \tilde{x}_1 \tag{76}$$

Because of the equations (74) and (76), and because the derivative of the high frequency signal $\tilde{x}_1$ is in $\frac{1}{\alpha}^2$, the derivative of x has the following expansion in α :

$$\frac{dx}{dt} = \left[\frac{d\bar{x}_0}{dt} + \frac{d\tilde{x}_0}{dt} + \alpha \frac{d\bar{x}_1}{dt} \right] + \alpha \frac{d\tilde{x}_1}{dt} \tag{77}$$

²The integral of HP is in α , so we consider that the derivative of HP is formally in $\frac{1}{\alpha}$

But, because x is the solution of the original dynamics equation (2), using the formal expansions in α of x and u , and thanks to assumption 2, we have another expansion of $\frac{dx}{dt}$ at the first order in α :

$$\frac{dx}{dt} = f(x_0, u_0, t) + \alpha(f_u(x_0, u_0, t)u_1 + f_x(x_0, u_0, t)x_1) \quad (78)$$

Consequently, identifying the zero order terms in equations (77) and (78), we obtain:

$$\frac{d\bar{x}_0}{dt} + \frac{d\tilde{x}_0}{dt} + \alpha \frac{d\tilde{x}_1}{dt} = f(x_0, u_0, t)$$

But by definition of $\bar{x}_0$ as the low frequency part of x_0 , we have:

$$\frac{d\bar{x}_0}{dt} = LP[f, u_0](x_0, t)$$

Consequently, by definition of HP , we have:

$$\frac{d\tilde{x}_0}{dt} + \alpha \frac{d\tilde{x}_1}{dt} = HP[f, u_0](x_0, t)$$

with initial value 0. Integration the previous equation yields

$$\tilde{x}_0 + \alpha\tilde{x}_1 = I[f, x_0, u_0]$$

But $I[f, \bar{x}_0, u_0]$ is of order 1 in α and $\tilde{x}_0$ is of order 0. Thus $\tilde{x}_0 = 0$, that gives $x_0 = \bar{x}_0$ the solution of the averaged problem. A consequence of that is that both definitions of x_0 and u_0 are consistent.

Moreover, we have a definition of $\tilde{x}_1$ consistent with definition 5:

$$\alpha\tilde{x}_1 = I[f, x_0, u_0]$$

so that so that the derivative of $\tilde{x}_1$ is $-\frac{1}{\alpha}HP[f_x, u_0](x_0, p_0, t)$, that is in $\frac{1}{\alpha}$.

C.2 Expansion of the costate p

The costate variable p is the solution of the costate dynamics equation with ending condition (29). We expand it at the first order with respect to α :

$$p = p_0 + \alpha p_1 \quad (79)$$

We then separate the expansion terms p_0 and p_1 into low (LP) and high (HP) frequencies components:

$$p_0 = \bar{p}_0 + \tilde{p}_0, \quad p_1 = \bar{p}_1 + \tilde{p}_1 \quad (80)$$

Because of the equations (79) and (80), and because the derivative of the high frequency signal $\tilde{p}_1$ is in $\frac{1}{\alpha}$, the derivative of p has the following expansion in α :

$$\frac{dp}{dt} = \left[\frac{d\bar{p}_0}{dt} + \frac{d\tilde{p}_0}{dt} + \alpha \frac{d\tilde{p}_1}{dt} \right] + \alpha \frac{d\bar{p}_1}{dt} \quad (81)$$

On the other hand, p is the solution of the costate dynamics equation with ending condition (29). Moreover, we have defined the developments in α of x , p and u .

Thus we have another development of $\frac{dp}{dt}$ at the first order in α :

$$\frac{dp}{dt} = -H_x(x_0, u_0, p_0, t) + \alpha(-H_{xu}(x_0, u_0, p_0, t)u_1 - H_{xx}(x_0, u_0, p_0, t)x_1 - f_x(x_0, u_0, t)p_1) \quad (82)$$

Consequently, identifying the zero order terms in equations (81) and (82), we have:

$$\frac{d\bar{p}_0}{dt} + \frac{d\tilde{p}_0}{dt} + \alpha \frac{d\tilde{p}_1}{dt} = -H_x(x_0, u_0, p_0, t)$$

But by definition of $\bar{p}_0$ as the low frequency part of p_0 , we have:

$$\frac{d\bar{p}_0}{dt} = -LP[H_x, u_0](x_0, p_0, t)$$

Consequently, by definition of HP , we have:

$$\frac{d\tilde{p}_0}{dt} + \alpha \frac{d\tilde{p}_1}{dt} = -HP[H_x, u_0](x_0, p_0, t)$$

with final value 0. Integration of the previous equation yields

$$\tilde{p}_0 + \alpha \tilde{p}_1 = I^T[H_x, (x_0, p_0), u_0]$$

But $I[-H_x, x_0, p_0, u_0]$ is of order 1 in α as $\tilde{p}_0$ is of order 0. Thus $\tilde{p}_0 = 0$, that gives $p_0 = \bar{p}_0$, the solution of the averaged costate equation. A consequence of that is that both definitions of p_0 are consistent.

Moreover, we have a definition of $\tilde{p}_1$ consistent with definition 5:

$$\alpha \tilde{p}_1 = I^T[H_x, (x_0, p_0), u_0]$$

so that the derivative of $\tilde{p}_1$ is $-\frac{1}{\alpha}HP[H_x, u_0](x_0, p_0, t)$, that is in $\frac{1}{\alpha}$.

D Proof of non negativity of $H_{\sigma\sigma}$

We make a proof by contradiction.

Let's suppose that (37) is not true. Then there exists a negative eigenvalue $-\gamma$ of $H_{\sigma\sigma}(x, u, p_0, t)$, that is there exists an eigenvector $\begin{bmatrix} y \\ v \end{bmatrix}$ so that:

$$H_{\sigma\sigma}(x, u, p_0, t) \begin{bmatrix} y \\ v \end{bmatrix} = -\gamma \begin{bmatrix} y \\ v \end{bmatrix}$$

This implies that the two following equations hold:

$$H_{xx}(x, u, p_0, t)y + H_{xu}(x, u, p_0, t)v = -\gamma y \quad (83)$$

$$H_{ux}(x, u, p_0, t)y + H_{uu}(x, u, p_0, t)v = -\gamma v \quad (84)$$

But $(\gamma Id + H_{uu}) \geq (\gamma + \beta)Id > 0$, so that it is invertible and the equation (84) can be solved in v , giving:

$$v = (\gamma Id + H_{uu}(x, u, p_0, t))^{-1} H_{ux}y \quad (85)$$

Then, replacing v by its value in the equation (83), we have:

$$(H_{xx}(x, u, p_0, t) - H_{xu}(x, u, p_0, t)(\gamma Id + H_{uu}(x, u, p_0, t))^{-1}H_{ux}(x, u, p_0, t))y = -\gamma y \quad (86)$$

But as $\gamma > 0$, we have the succession of inequalities:

$$(\gamma Id + H_{uu}(x, u, p_0, t)) \geq H_{uu}(x, u, p_0, t)$$

then

$$(\gamma Id + H_{uu}(x, u, p_0, t))^{-1} \leq H_{uu}^{-1}(x, u, p_0, t)$$

and then

$$\begin{aligned} (H_{xx}(x, u, p_0, t) - H_{xu}(x, u, p_0, t)(\gamma Id + H_{uu}(x, u, p_0, t))^{-1}H_{ux}(x, u, p_0, t)) \\ \geq [H_{xx} - H_{xu}H_{uu}^{-1}H_{ux}](x, u, p_0, t) \geq 0 \end{aligned}$$

Thus the equation (86) can not hold, because $-\gamma < 0$ can not be an eigenvalue.

The Proposition 7 is thus proved by contradiction.

E Proof of the inequalities on r , v , $r - r_1$ and r_1

E.1 Proof of proposition 12

Proposition 12 is important since it bounds r and v with respect to z and α . To obtain a bound on r and v which only depends on α , it will be enough to provide a bound of z involving α only.

E.1.1 Upper bound for r

Proposition 22. *The dynamics of r is the following:*

$$\begin{aligned} \frac{dr}{dt} &= f(r + x_0 + \alpha(\tilde{x}_1 + y_1), v + u_0 + \alpha v_1, t) - f(x_0 + \alpha(\tilde{x}_1 + y_1), u_0 + \alpha v_1, t) \\ &\quad + \alpha^2 \int_0^1 \int_0^1 \lambda f_{\sigma\sigma}(x_0 + \lambda\mu\alpha(\tilde{x}_1 + y_1), v + u_0 + \alpha v_1, t) \begin{bmatrix} \tilde{x}_1 + y_1 \\ v_1 \end{bmatrix}^2 d\lambda d\mu \end{aligned} \quad (87)$$

Proof. By definition of r and v , we have:

$$\begin{aligned} r &= x - x_0 - \alpha(\tilde{x}_1 + y_1), \quad \text{so that : } x = r + x_0 + \alpha(\tilde{x}_1 + y_1) \\ v &= u - u_0 - v_1, \quad \text{so that : } u = v + u_0 + v_1 \end{aligned}$$

Moreover:

$$\begin{aligned} \frac{dr}{dt} &= \frac{dx}{dt} - \frac{dx_0}{dt} - \alpha \frac{d\tilde{x}_1}{dt} - \alpha \frac{dy_1}{dt} \\ &= f(r + x_0 + \alpha(\tilde{x}_1 + y_1), v + u_0 + v_1, t) - LP[f, u_0](x_0, t) \\ &\quad - HP[f, u_0](x_0, t) - \alpha[f_{0x}(y_1 + \tilde{x}_1) + f_{0u}v_1] \\ &= f(r + x_0 + \alpha(\tilde{x}_1 + y_1), v + u_0 + v_1, t) - f(x_0, u_0, t) \\ &\quad - \alpha[f_{0x}(y_1 + \tilde{x}_1) + f_{0u}v_1] \end{aligned} \quad (88)$$

But a Taylor expansion of $f(x_0 + \alpha(\tilde{x}_1 + y_1), u_0 + \alpha v_1, t)$ is so:

$$\begin{aligned} f(x_0(\tilde{x}_1 + y_1), u_0 + \alpha v_1, t) &= f(x_0, u_0, t) + \alpha[f_{0x}(y_1 + \tilde{x}_1) + f_{0u}v_1] \\ &+ \alpha^2 \int_0^1 \int_0^1 \lambda f_{\sigma\sigma}(x_0 + \lambda\mu\alpha(x_1 + \tilde{y}_1, v + u_0 + \alpha v_1, t) \begin{bmatrix} \tilde{x}_1 + y_1 \\ v_1 \end{bmatrix})^2 d\lambda d\mu \end{aligned} \quad (89)$$

Introducing equation (89) in equation (88) proves equation (87). $\square$

Proposition 23. *The following inequality holds:*

$$\|r\|_\infty \leq k e^{k(1+\frac{K}{\beta})} \left[\sqrt{T} \|Z(\lambda, \mu)\|_2 + T \left(\frac{K}{\beta} + \frac{1}{2}((1+2M)^2\alpha) \right) \alpha \right] \quad (90)$$

Proof. Equation (33) about the upper bound of $f_{\sigma\sigma}$ leads to:

$$\begin{aligned} \left| f_{\sigma\sigma}(x_0 + \lambda\mu\alpha(x_1 + \tilde{y}_1, v + u_0 + \alpha v_1, t) \begin{bmatrix} \tilde{x}_1 + y_1 \\ v_1 \end{bmatrix})^2 \right| &\leq k(|\tilde{x}_1 + y_1| + |v_1|)^2 \\ &\leq k(|\tilde{x}_1| + |y_1| + |v_1|)^2 \end{aligned}$$

Hence:

$$\left| f_{\sigma\sigma}(x_0 + \lambda\mu\alpha(x_1 + \tilde{y}_1, v + u_0 + \alpha v_1, t) \begin{bmatrix} \tilde{x}_1 + y_1 \\ v_1 \end{bmatrix})^2 \right| \leq k(1+2M)^2 \quad (91)$$

Moreover, the function f is Lipschitz in x and u with Lipschitz constant the bound of the derivatives k , so that:

$$|f(r+x_0+\alpha(\tilde{x}_1+y_1), v+u_0+\alpha v_1, t) - f(x_0+\alpha(\tilde{x}_1+y_1), u_0+\alpha v_1, t)| \leq k(|r|+|v|) \quad (92)$$

The equations (91) and (92) in the equation (87) give, together with the fact that $r(0) = 0$ the following inequality:

$$|r(t)| \leq \int_0^t (|r(s)| + |v(s)|) ds + \frac{kT}{2} (2M+1)^2 \alpha^2 \quad (93)$$

But by definition of $Z[\lambda, \mu]$, we have:

$$v = Z[\lambda, \mu] - [H_{uu}^{-1} H_{ux}](\rho(\lambda, \mu), p_0, t)(r + \alpha \tilde{x}_1) \quad (94)$$

so that:

$$|v(s)| \leq |Z[\lambda, \mu](s)| + \|H_{uu}^{-1}\|_\infty \|H_{ux}\|_\infty (|r(s)| + \alpha |\tilde{x}_1(s)|)$$

But the assumption 3 proves that H_{uu} is invertible and that H_{uu}^{-1} is bounded by $\frac{1}{\beta}$, so that:

$$|v(s)| \leq |Z[\lambda, \mu](s)| + \frac{K}{\beta} (|r(s)| + \alpha) \quad (95)$$

Including equation(95) in equation (93) leads to:

$$|r(t)| \leq k \left(1 + \frac{K}{\beta} \right) \int_0^t (|r(s)|) ds + \frac{kKT}{\beta} \alpha + \frac{kT}{2} (2M+1)^2 \alpha^2 + k \int_0^T |Z[\lambda, \mu](t)| dt \quad (96)$$

But Cauchy property leads to:

$$\int_0^T |Z[\lambda, \mu](t)| dt \leq \sqrt{T} \|Z(\lambda, \mu)\|_2 \quad (97)$$

Including equation (97) in equation (96) lead to:

$$|r(t)| \leq k \left(1 + \frac{K}{\beta}\right) \int_0^t (|r(s)|) ds + k\sqrt{T} \|Z(\lambda, \mu)\|_2 + kT \left[\frac{K}{\beta} + \frac{1}{2}(2M+1)^2 \alpha \right] \alpha \quad (98)$$

Equation (98), together with Gronwall lemma, proves equation (90). $\square$

Proof of Equation (44) Let's take the square of equation (90):

$$\|r\|_\infty^2 \leq 2k^2 e^{2k(1+\frac{K}{\beta})} \left[T \|Z(\lambda, \mu)\|_2^2 + T^2 \left(\frac{K}{\beta} + \frac{1}{2}((1+2M)^2 \alpha) \right)^2 \alpha^2 \right]$$

Let's now multiply by λ and integrate relatively to λ and μ between 0 and 1:

$$\frac{1}{2} \|r\|_\infty^2 \leq 2k^2 e^{2k(1+\frac{K}{\beta})} \left[T z^2 + \frac{1}{2} T^2 \left(\frac{K}{\beta} + \frac{1}{2}((1+2M)^2 \alpha) \right)^2 \alpha^2 \right]$$

Multiplying by 2 that equation proves Equation (44).

E.1.2 Upper bound for v

Let's apply the triangular inequality for the $\mathbf{L}_2$ norm to the expression of v (94):

$$\|v\|_2 \leq \|Z[\lambda, \mu]\|_2 + \|[H_{uu}^{-1} H_{ux}](\rho(\lambda, \mu), p_0, t)(r + \alpha \tilde{x}_1)\|_2$$

But:

$$\|[H_{uu}^{-1} H_{ux}](\rho(\lambda, \mu), p_0, t)(r + \alpha \tilde{x}_1)\|_2 \leq \sqrt{T} \|[H_{uu}^{-1} H_{ux}](\rho(\lambda, \mu), p_0, t)(r + \alpha \tilde{x}_1)\|_\infty \leq \frac{K\sqrt{T}}{\beta} (\|r\|_\infty + \alpha \|\tilde{x}_1\|_\infty)$$

Thus, taking the squares:

$$\|v\|_2^2 \leq 3 \left[\|Z[\lambda, \mu]\|_2^2 + \frac{K^2 T}{\beta^2} (\|r\|_\infty^2 + \alpha^2 \|\tilde{x}_1\|_\infty^2) \right]$$

Let's multiply by λ and integrate relatively to λ and μ between 0 and 1:

$$\frac{1}{2} \|v\|_2^2 \leq 3 \left[z^2 + \frac{K^2 T}{2\beta^2} (\|r\|_\infty^2 + \alpha^2) \right] \leq 3 \left[\left(1 + \frac{K^2 T k_{r1}}{2\beta^2} \right) z^2 + \frac{K^2 T}{2\beta^2} (1 + k_{r2}) \alpha^2 \right]$$

Multiplying by 2 that equation proves Equation (45).

E.2 Proof of proposition 13

We now turn our attention to the variable r , which can be approximated by the variable r_1 , up to an error which is bounded by a term that depends only on z and α .

The variable r follows the dynamic (87) with $r(0) = 0$, and r_1 follows the dynamic (46).

Thus $r - r_1$ follows the dynamics:

$$\begin{aligned} \frac{d(r - r_1)}{dt} &= f(r + x_0 + \alpha(\tilde{x}_1 + y_1), v + u_0 + \alpha v_1, t) - f(r_1 + x_0 + \alpha(\tilde{x}_1 + y_1), v + u_0 + \alpha v_1, t) \\ &\quad + \alpha^2 \int_0^1 \int_0^1 \lambda f_{\sigma\sigma}(x_0 + \lambda\mu\alpha(\tilde{x}_1 + y_1), v + u_0 + \alpha v_1, t) \begin{bmatrix} \tilde{x}_1 + y_1 \\ v_1 \end{bmatrix}^2 d\lambda d\mu \end{aligned}$$

Thus, in a similar way than for the upper bound of r , the following inequality holds:

$$|r(t) - r_1(t)| \leq k \int_0^t |r(s) - r_1(s)| ds + \frac{kT}{2} (2M + 1)^2 \alpha^2$$

The equation (47) follows from Gronwall lemma.

The equation (48) is then the consequence of equations (44) and (47), together with:

$$\|r_1\|_\infty^2 \leq 2(\|r\|_\infty^2 + \|r - r_1\|_\infty^2)$$

F Proof of the expansion of the real cost (proposition 14)

Proposition 24. $L(x, u, t)$ expands the following way:

$$L(x, u, t) = L(x_0, u_0, t) + L_x(x_0, u_0, t)\delta x + L_u(x_0, u_0, t)\delta u + \int_0^1 \int_0^1 \lambda L_{\sigma\sigma}(\rho(\lambda, \mu), t)(\delta\sigma)^2 d\lambda d\mu \quad (99)$$

Proof. It is a Taylor expansion of $L(x, u, t)$ with integral remainder. $\square$

Proposition 25. The dynamics $\frac{d\tilde{x}}{dt}$ of $\tilde{x}$ expands the following way:

$$\frac{d\tilde{x}}{dt} = f_x(x_0, u_0, t)\delta x + f_u(x_0, u_0, t)\delta u + \int_0^1 \int_0^1 \lambda f_{\sigma\sigma}(\rho(\lambda, \mu), t)(\delta\sigma)^2 d\lambda d\mu \quad (100)$$

Proof. By definition of $\tilde{x}$, we have

$$\tilde{x} = x - x_0 - \alpha\tilde{x}_1,$$

so that:

$$\begin{aligned} \frac{d\tilde{x}}{dt} &= f(x_0 + \delta x, u_0 + \delta u, t) - LP[f, u_0](x_0, t) + HP[f, u_0](x_0, t) \\ &= f(x_0 + \delta x, u_0 + \delta u, t) - f(x_0, u_0, t) \end{aligned}$$

The equation (100) follows by Taylor expansion of $f(x, u, t)$ with integral remainder. $\square$

Proposition 26. $L(x, u, t)$ rewrites the following way:

$$L(x, u, t) = L(x_0, u_0, t) + H_{0x}(\tilde{x} + \alpha\tilde{x}_1) - p_0 \frac{d\tilde{x}}{dt} + \int_0^1 \int_0^1 \lambda H_{\sigma\sigma}(\rho(\lambda, \mu), p_0, t)(\delta\sigma)^2 d\lambda d\mu \quad (101)$$

Proof. We have the following identities:

$$\delta x = \tilde{x} + \alpha \tilde{x}_1$$

and:

$$L_x(x_0, u_0, t) = H_{0x} - p_0 f_x(x_0, u_0, t)$$

Moreover, because of the stationary condition of the averaged problem, we have:

$$L_u(x_0, u_0, t) = -p_0 f_u(x_0, u_0, t)$$

Finally we change the integral remainder of the expansion of $L(x, u, t)$ in equation (99) with:

$$L_{\sigma\sigma}(\rho(\lambda, \mu), t) = H_{\sigma\sigma}(\rho(\lambda, \mu), p_0, t) - p_0 f_{\sigma\sigma}(\rho(\lambda, \mu), t)$$

Thus equation (99) leads to:

$$\begin{aligned} L(x, u, t) &= L(x_0, u_0, t + H_{0x}(\tilde{x} + \alpha \tilde{x}_1) - p_0 [f_x(x_0, u_0, t) + f_u(x_0, u_0, t) \\ &+ \int_0^1 \int_0^1 \lambda f_{\sigma\sigma}(\rho(\lambda, \mu), t)] (\delta\sigma)^2 d\lambda d\mu] + \int_0^1 \int_0^1 \lambda H_{\sigma\sigma}(\rho(\lambda, \mu), p_0, t) (\delta\sigma)^2 d\lambda d\mu \end{aligned}$$

Inserting equation (100) in this equation proves proposition 26. $\square$

Proposition 27. *The following equality holds:*

$$\int_0^T -p_0 \frac{d\tilde{x}}{dt} dt = - \int_0^T LP[H_x, u_0](x_0, t) \tilde{x} dt \quad (102)$$

Proof. Let's make an integration by part:

$$\int_0^T -p_0 \frac{d\tilde{x}}{dt} dt = -[p_0 \tilde{x}]_0^T + \int_0^T \frac{dp_0}{dt} \tilde{x} dt$$

This leads to the equation (102) because $\tilde{x}(0) = 0$, $p_0(T) = 0$ and:

$$\frac{dp_0}{dt} = -LP[H_x, u_0](x_0, t)$$

$\square$

Now inserting the equation (102) in the equation (101) integrated between 0 and T leads to equation (52), which ends the proof of proposition 14.

Remark The remaining appendix sections are devoted to the obtention of a lower bound on $J(u)$. This is achieved by obtaining a lower bound on each term of the expansion (52) of $J(u)$. Only the complicated terms require an extensive study; this study is detailed in sections G, H, I and finally J.

G Proof of the lower bound of the third term (Proposition 15)

Proposition 28. *The following inequality holds:*

$$\int_0^T HP[H_x, u_0](x_0, p_0, t) \tilde{x} dt \geq -(2KTk_{r5} + k(1 + 2M))\alpha^2 + \alpha \int_0^T \tilde{p}_1 \frac{dr_1}{dt} dt \quad (103)$$

Proof. By definition of r , we have:

$$\tilde{x} = r + \alpha y_1 = (r - r_1) + (r_1 + \alpha y_1)$$

So that

$$\int_0^T HP[H_x, u_0](x_0, p_0, t) \tilde{x} dt = \int_0^T HP[H_x, u_0](x_0, p_0, t)(r - r_1) dt + \int_0^T HP[H_x, u_0](x_0, p_0, t)(r_1 + \alpha y_1) dt$$

But because of equation (47), we have

$$\left| \int_0^T HP[H_x, u_0](x_0, p_0, t)(r - r_1) dt \right| \leq T \|HP[H_x, u_0]\|_\infty k_{r5} \alpha^2 \leq 2KTk_{r5} \alpha^2$$

Moreover, by definition of $\alpha \tilde{p}_1$, we have:

$$HP[H_x, u_0](x_0, p_0, t) = -\alpha \frac{d\tilde{p}_1}{dt}(t)$$

So that the following inequality holds:

$$\int_0^T HP[H_x, u_0](x_0, p_0, t) \tilde{x} dt \geq -2KTk_{r5} \alpha^2 - \alpha \int_0^T \frac{d\tilde{p}_1}{dt} (\tilde{x}_1 + \alpha y_1) dt$$

If we make an integration by part and use the fact that $\tilde{p}_1(T) = \tilde{x}_1(0) = y_1(0) = 0$, we get:

$$\int_0^T HP[H_x, u_0](x_0, p_0, t) \tilde{x} dt \geq -2KTk_{r5} \alpha^2 + \alpha \int_0^T \tilde{p}_1 \left(\frac{d\tilde{x}_1}{dt} + \alpha \frac{dy_1}{dt} \right) dt$$

y_1 follows the dynamics (40) and $\|\tilde{p}_1\|_\infty \leq 1$, so that:

$$\left| \alpha^2 \int_0^T \tilde{p}_1 \frac{dy_1}{dt} dt \right| \leq k\alpha^2 (\|y_1 + \tilde{x}_1\|_\infty + \|v_1\|_\infty) \leq k(2M + 1)\alpha^2$$

So that the inequality (103) is proved. $\square$

Proposition 29. *The dynamics of r_1 expands the following way:*

$$\begin{aligned} \frac{dr_1}{dt} = & [f_{0x}r_1 + f_{0u}v] \\ & + \int_0^1 \int_0^1 \lambda f_{\sigma\sigma}(x_0 + \lambda\mu(r_1 + \alpha(\tilde{x}_1 + y_1)), u_0 + \lambda\mu(v + \alpha v_1), t) \left[\begin{array}{c} r_1 + \alpha(\tilde{x}_1 + y_1) \\ v + \alpha v_1 \end{array} \right]^2 d\lambda d\mu \\ & - \alpha^2 \int_0^1 \int_0^1 \lambda f_{\sigma\sigma}(x_0 + \lambda\mu\alpha(\tilde{x}_1 + y_1), u_0 + \lambda\mu\alpha v_1, t) \left[\begin{array}{c} \tilde{x}_1 + y_1 \\ \alpha v_1 \end{array} \right]^2 d\lambda d\mu \end{aligned} \quad (104)$$

Proof. The dynamics of r_1 (equation (46)) is the difference between the quantities $f(r_1 + x_0 + \alpha(x_1 + y_1), v + u_0 + \alpha v_1, t)$ and $f(x_0 + \alpha(x_1 + y_1), u_0 + \alpha v_1, t)$.

Let's make the Taylor expansions of these quantities at (x_0, u_0) :

$$\begin{aligned} f(r_1 + x_0 + \alpha(x_1 + y_1), v + u_0 + \alpha v_1, t) \\ = f(x_0, u_0, t) \\ + f_{0x}(r_1 + \alpha(x_1 + y_1)) + f_{0u}(v + \alpha v_1) \\ + \int_0^1 \int_0^1 \lambda f_{\sigma\sigma}(x_0 + \lambda\mu(r_1 + \alpha(\tilde{x}_1 + y_1)), u_0 + \lambda\mu(v + \alpha v_1), t) \left[\frac{r_1 + \alpha(\tilde{x}_1 + y_1)}{v + \alpha v_1} \right]^2 d\lambda d\mu \end{aligned}$$

and

$$\begin{aligned} f(x_0 + \alpha(x_1 + y_1), u_0 + \alpha v_1, t) \\ = f(x_0, u_0, t) \\ + \alpha[f_{0x}(x_1 + y_1) + f_{0u}v_1] \\ + \alpha^2 \int_0^1 \int_0^1 \lambda f_{\sigma\sigma}(x_0 + \lambda\mu\alpha(\tilde{x}_1 + y_1), u_0 + \lambda\mu\alpha v_1, t) \left[\frac{\tilde{x}_1 + y_1}{v_1} \right]^2 d\lambda d\mu \end{aligned}$$

The simplifications between the two expansions while we take their differences gives the dynamics of r_1 (104). $\square$

The consequence of that dynamics expansion is that:

$$\begin{aligned} \alpha \int_0^T \tilde{p}_1 \frac{dr_1}{dt} dt &= \alpha \int_0^T \tilde{p}_1 [f_{0x}r_1 + f_{0u}v] dt \\ &+ \alpha \int_0^T \tilde{p}_1 \left[\int_0^1 \int_0^1 \lambda f_{\sigma\sigma}(x_0 + \lambda\mu(r_1 + \alpha(\tilde{x}_1 + y_1)), u_0 + \lambda\mu(v + \alpha v_1), t) \left[\frac{r_1 + \alpha(\tilde{x}_1 + y_1)}{v + \alpha v_1} \right]^2 d\lambda d\mu \right] dt \\ &- \alpha^3 \int_0^T \tilde{p}_1 \left[\int_0^1 \int_0^1 \lambda f_{\sigma\sigma}(x_0 + \lambda\mu\alpha(\tilde{x}_1 + y_1), u_0 + \lambda\mu\alpha v_1, t) \left[\frac{\tilde{x}_1 + y_1}{v_1} \right]^2 d\lambda d\mu \right] dt \end{aligned} \quad (105)$$

Proposition 30. *Let's define the terms:*

$$R_1 = \alpha \int_0^T \tilde{p}_1 \left[\int_0^1 \int_0^1 \lambda f_{\sigma\sigma}(x_0 + \lambda\mu(r_1 + \alpha(\tilde{x}_1 + y_1)), u_0 + \lambda\mu(v + \alpha v_1), t) \left[\frac{r_1 + \alpha(\tilde{x}_1 + y_1)}{v + \alpha v_1} \right]^2 d\lambda d\mu \right] dt$$

and

$$R_2 = \alpha^3 \int_0^T \tilde{p}_1 \left[\int_0^1 \int_0^1 \lambda f_{\sigma\sigma}(x_0 + \lambda\mu\alpha(\tilde{x}_1 + y_1), u_0 + \lambda\mu\alpha v_1, t) \left[\frac{\tilde{x}_1 + y_1}{v_1} \right]^2 d\lambda d\mu \right] dt$$

Then the following inequalities hold:

$$|R_1| \leq \frac{3k}{2} [(k_{r3}T + k_{v1})\alpha z^2 + [(k_{r4}T + k_{v2} + T(1 + 2M)^2)\alpha^3] \quad (106)$$

and

$$|R_2| \leq \frac{kT}{2} (1 + 2M)^2 \alpha^3 \quad (107)$$

Proof. Equation (33) about the upper bound of $f_{\sigma\sigma}$ leads to:

$$\begin{aligned}
& \left| f_{\sigma\sigma}(x_0 + \lambda\mu\alpha(\tilde{x}_1 + y_1), u_0 + \lambda\mu\alpha v_1, t) \begin{bmatrix} r_1 + \alpha(\tilde{x}_1 + y_1) \\ v + \alpha v_1 \end{bmatrix}^2 \right| \\
& \leq 3k[|r_1 + \alpha(\tilde{x}_1 + y_1)| + |v + \alpha v_1|]^2 \\
& \leq 3k[|r_1| + |v| + \alpha(|\tilde{x}_1 + y_1| + |v_1|)]^2 \\
& \leq 3k[r_1^2 + v^2 + \alpha(|\tilde{x}_1 + y_1| + |v_1|)^2]
\end{aligned}$$

Thus, together with the fact that $\tilde{p}_1 \leq 1$, we have:

$$\begin{aligned}
|R_1| & \leq \frac{3k\alpha}{2}[\|r_1\|_2^2 + \|v\|_2^2 + \alpha^2 T(2M+1)^2] \\
|R_1| & \leq \frac{3k\alpha}{2}[\|r_1\|_\infty^2 + \|v\|_2^2 + \alpha^2 T(2M+1)^2]
\end{aligned}$$

Equations (48) and (45) then lead to equation (106).

Equation (33) about the upper bound of $f_{\sigma\sigma}$ leads also to:

$$\left| f_{\sigma\sigma}(x_0 + \lambda\mu\alpha(\tilde{x}_1 + y_1), u_0 + \lambda\mu\alpha v_1, t) \begin{bmatrix} r_1 + \alpha(\tilde{x}_1 + y_1) \\ v + \alpha v_1 \end{bmatrix}^2 \right| \leq k(|\tilde{x}_1 + y_1| + |v_1|)^2 \leq k(1+2M)^2$$

That proves equation (107) by triple integration and multiplication by α^3 . $\square$

The equations (106) and (107) included in equation (107) (105) lead to:

$$\begin{aligned}
\alpha \int_0^T \tilde{p}_1 \frac{dr_1}{dt} dt &= \alpha \int_0^T \tilde{p}_1 [f_{0x} r_1 + f_{0u} v] dt \\
&\quad - \frac{3k}{2} (k_{r2} T + k_{v1}) M \alpha z^2 - \frac{k}{2} [3(k_{r4} T + k_{v2}) + 4T(1+2M)^2] \alpha^3
\end{aligned}$$

Introducing that equation into equation (103) leads to equation (53) and thus the proposition 15.

H Proof of the lower bound of the fourth term (Proposition 16)

Proposition 31. *The following inequality holds:*

$$\begin{aligned}
& \int_0^T \int_0^1 \int_0^1 \lambda H_{\sigma\sigma}(\rho(\lambda, \mu), p_0, t) (\delta\sigma)^2 d\lambda d\mu dt \\
& \geq -2KT M^2 \alpha^2 + \beta z^2 \\
& + 2\alpha \int_0^T \int_0^1 \int_0^1 \lambda [y_1, v_1] H_{\sigma\sigma}(\rho(\lambda, \mu), p_0 t) \begin{bmatrix} r + \alpha \tilde{x}_1 \\ v \end{bmatrix} d\lambda d\mu dt
\end{aligned} \tag{108}$$

Proof. Let's use the fact that:

$$\delta\sigma = \begin{bmatrix} r + \alpha \tilde{x}_1 \\ v \end{bmatrix} + \alpha \begin{bmatrix} y_1 \\ v_1 \end{bmatrix}$$

to expand $H_{\sigma\sigma}(\rho(\lambda, \mu), p_0, t)(\delta\sigma)^2$:

$$\begin{aligned} H_{\sigma\sigma}(\rho(\lambda, \mu), p_0, t)(\delta\sigma)^2 &= H_{\sigma\sigma}(\rho(\lambda, \mu), p_0, t) \left[\begin{array}{c} r + \alpha\tilde{x}_1 \\ v \end{array} \right]^2 \\ &\quad + \alpha^2 H_{\sigma\sigma}(\rho(\lambda, \mu), p_0, t) \left[\begin{array}{c} y_1 \\ v_1 \end{array} \right]^2 \\ &\quad + 2\alpha[y_1, v_1] H_{\sigma\sigma}(\rho(\lambda, \mu), p_0, t) \left[\begin{array}{c} r + \alpha\tilde{x}_1 \\ v \end{array} \right] \end{aligned} \quad (109)$$

The second term is easily upper bounded in absolute value (equation (34) about the upper bound of $H_{\sigma\sigma}$):

$$\left| H_{\sigma\sigma}(\rho(\lambda, \mu), p_0, t) \left[\begin{array}{c} y_1 \\ v_1 \end{array} \right]^2 \right| \leq K(|y_1| + |v_1|)^2 \leq 4KM^2$$

so that:

$$\left| \int_0^T \int_0^1 \int_0^1 H_{\sigma\sigma}(\rho(\lambda, \mu), p_0, t) \left[\begin{array}{c} y_1 \\ v_1 \end{array} \right]^2 d\lambda d\mu dt \right| \leq 2KTM^2 \quad (110)$$

To upper bound the first term, let's expand it in its components:

$$\begin{aligned} H_{\sigma\sigma}(\rho(\lambda, \mu), p_0, t) \left[\begin{array}{c} r + \alpha\tilde{x}_1 \\ v \end{array} \right]^2 &= H_{xx}(\rho(\lambda, \mu), p_0, t)(r + \alpha\tilde{x}_1)^2 \\ &\quad + 2(r + \alpha\tilde{x}_1)H_{xu}(\rho(\lambda, \mu), p_0, t)v \\ &\quad + H_{uu}(\rho(\lambda, \mu), p_0, t)v^2 \end{aligned} \quad (111)$$

Now let's use the definition of $Z[\lambda, \mu](t) = v + [H_{uu}^{-1}H_{ux}](\rho(\lambda, \mu), p_0, t)(r + \alpha\tilde{x}_1)$ to expand the terms in v and v^2 :

$$\begin{aligned} 2(r + \alpha\tilde{x}_1)H_{xu}(\rho(\lambda, \mu), p_0, t)v &= 2(r + \alpha\tilde{x}_1)H_{xu}(\rho(\lambda, \mu), p_0, t)Z[\lambda, \mu] \\ &\quad - 2[H_{ux}H_{uu}^{-1}H_{xu}](\rho(\lambda, \mu), p_0, t)(r + \alpha\tilde{x}_1)^2 \end{aligned} \quad (112)$$

and

$$\begin{aligned} H_{uu}(\rho(\lambda, \mu), p_0, t)v^2 &= H_{uu}(\rho(\lambda, \mu), p_0, t)Z[\lambda, \mu]^2 \\ &\quad + [H_{ux}H_{uu}^{-1}H_{xu}](\rho(\lambda, \mu), p_0, t)(r + \alpha\tilde{x}_1)^2 \\ &\quad - 2(r + \alpha\tilde{x}_1)H_{xu}(\rho(\lambda, \mu), p_0, t)Z[\lambda, \mu] \end{aligned} \quad (113)$$

Introducing equations (112) and (113) in equation (111) leads to:

$$\begin{aligned} H_{\sigma\sigma}(\rho(\lambda, \mu), p_0, t) \left[\begin{array}{c} r + \alpha\tilde{x}_1 \\ v \end{array} \right]^2 &= [H_{xx} - H_{ux}H_{uu}^{-1}H_{xu}](\rho(\lambda, \mu), p_0, t)(r + \alpha\tilde{x}_1)^2 \\ &\quad + H_{uu}Z[\lambda, \mu]^2 \end{aligned}$$

Because of assumption 3 on the convexity, this proves that:

$$H_{\sigma\sigma}(\rho(\lambda, \mu), p_0, t) \left[\begin{array}{c} r + \alpha\tilde{x}_1 \\ v \end{array} \right]^2 \geq \beta Z[\lambda, \mu]^2$$

so that:

$$\int_0^T \int_0^1 \int_0^1 H_{\sigma\sigma}(\rho(\lambda, \mu), p_0, t) \begin{bmatrix} r + \alpha\tilde{x}_1 \\ v \end{bmatrix}^2 d\lambda d\mu dt \geq \beta z^2 \quad (114)$$

Including equations (114) and (110) into equation (109) leads to equation (108). $\square$

Proposition 32. *Let's denote:*

$$J_{41} = 2\alpha \int_0^T \int_0^1 \int_0^1 \lambda [y_1, v_1] [H_{\sigma\sigma}(\rho(\lambda, \mu), p_0 t) - H_{\sigma\sigma}(w_0, t)] \begin{bmatrix} r + \alpha\tilde{x}_1 \\ v \end{bmatrix} d\lambda d\mu dt$$

Then the following bound holds:

$$|J_{41}| \leq 4\sqrt{3}KM\alpha \left[\left((Tk_{r1} + \frac{1}{2}k_{v1}) z^2 + \left(Tk_{r2} + \frac{1}{2}k_{v2} + \frac{T}{2}(M^2 + 2) \right) \alpha^2 \right) \right] \quad (115)$$

Proof. Let's expand $[y_1, v_1] [H_{\sigma\sigma}(\rho(\lambda, \mu), p_0 t) - H_{\sigma\sigma}(w_0, t)] \begin{bmatrix} r + \alpha\tilde{x}_1 \\ v \end{bmatrix}$ into its coordinates:

$$\begin{aligned} & [y_1, v_1] [H_{\sigma\sigma}(\rho(\lambda, \mu), p_0 t) - H_{\sigma\sigma}(w_0, t)] \begin{bmatrix} r + \alpha\tilde{x}_1 \\ v \end{bmatrix} \\ &= y_1 [H_{xx}(\rho(\lambda, \mu), p_0 t) - H_{xx}(w_0, t)] (r + \alpha\tilde{x}_1) \\ &+ y_1 [H_{xu}(\rho(\lambda, \mu), p_0 t) - H_{xu}(w_0, t)] v \\ &+ v_1 [H_{ux}(\rho(\lambda, \mu), p_0 t) - H_{ux}(w_0, t)] (r + \alpha\tilde{x}_1) \\ &+ v_1 [H_{uu}(\rho(\lambda, \mu), p_0 t) - H_{uu}(w_0, t)] v \end{aligned} \quad (116)$$

But the second derivatives of $H(x, u, p_0, t)$ are Lipschitz in (x, u) of Lipschitz constant K because the third derivatives of $H(x, u, p_0, t)$ are bounded by K . Thus with the definition of $\rho(\lambda, \mu)$ and w_0 , equation (116) leads to:

$$\begin{aligned} & |[y_1, v_1] [H_{\sigma\sigma}(\rho(\lambda, \mu), p_0 t) - H_{\sigma\sigma}(w_0, t)] \begin{bmatrix} r + \alpha\tilde{x}_1 \\ v \end{bmatrix}| \\ &\leq \lambda \mu K \sqrt{(r + \alpha\tilde{x}_1)^2 + \alpha^2 v_1^2} (|y_1| |r + \alpha\tilde{x}_1| + |y_1| |v| + |v| |r + \alpha\tilde{x}_1| + |v_1| |v|) \\ &\leq K \sqrt{(r + \alpha)^2 + \alpha^2 M^2} (2M(|r| + \alpha) + 2M|v|) \\ &\leq 2MK \sqrt{2r^2 + 2\alpha^2 + M^2 \alpha^2} \sqrt{3(r^2 + v^2 + \alpha^2)} \\ &\leq 2MK \sqrt{3} (2r^2 + v^2 + (M^2 + 2)\alpha^2) \end{aligned}$$

Thus, with a triple integration after multiplication by λ , we get:

$$|J_{41}| \leq 4\sqrt{3}KM\alpha \left[T\|r\|_\infty^2 + \frac{T}{2}\|v\|_2^2 + ((M^2 + 2)\alpha^2) \right]$$

Then, using equations (44) and (45) lead to equation (114). $\square$

Applying propositions 31 and 32 lead to:

$$\begin{aligned}
& \int_0^T \int_0^1 \int_0^1 \lambda H_{\sigma\sigma}(\rho(\lambda, \mu), p_0, t) (\delta\sigma)^2 d\lambda d\mu dt \\
& \geq -2KM \left[TM + \sqrt{3}(2Tk_{r2} + k_{v2} + T(M^2 + 2))\alpha \right] \alpha^2 \\
& + \left[\beta - 2\sqrt{3}KM(2Tk_{r1} + k_{v1})\alpha \right] z^2 \\
& + 2\alpha \int_0^T \int_0^1 \int_0^1 \lambda [y_1, v_1] H_{\sigma\sigma}(w_0, t) \begin{bmatrix} r + \alpha\tilde{x}_1 \\ v \end{bmatrix} d\lambda d\mu dt
\end{aligned}$$

That proves proposition 16 because $H_{0\sigma\sigma} = H_{\sigma\sigma}(w_0, t)$ does not depend on λ and μ .

I Proof of Bound in absolute value for the sum of the integral terms of the right hand sides of equations (53) and (54)

Thanks to the fact that $r + \alpha\tilde{x}_1 = r_1 + ((r - r_1) + \alpha\tilde{x}_1)$, we have $R = R_3 + R_4$, where R_3 and R_4 are defined as:

$$R_3 = \alpha \int_0^T \tilde{p}_1 \left[f_{0x}r_1 + f_{0u}v + [y_1, v_1] H_{0\sigma\sigma} \begin{bmatrix} r_1 \\ v \end{bmatrix} \right] dt$$

and

$$R_4 = \alpha \int_0^T [y_1, v_1] H_{0\sigma\sigma} \begin{bmatrix} r_1 + \alpha\tilde{x}_1 \\ 0 \end{bmatrix} dt$$

Proposition 33. *The following inequality holds:*

$$|R_3| \leq kM ((Tk_{r3} + k_{v1})\alpha z^2 + (Tk_{r4} + k_{v2})\alpha^3) \quad (117)$$

Proof. Let's develop $[y_1, v_1] H_{0\sigma\sigma} \begin{bmatrix} r_1 \\ v \end{bmatrix}$ into coordinates:

$$[y_1, v_1] H_{0\sigma\sigma} \begin{bmatrix} r_1 \\ v \end{bmatrix} = H_{0xx}y_1r_1 + H_{0ux}y_1v + H_{0xu}v_1r_1 + H_{0uu}v_1v$$

so that:

$$\begin{aligned}
& \tilde{p}_1 \left[f_{0x}r_1 + f_{0u}v + [y_1, v_1] H_{0\sigma\sigma} \begin{bmatrix} r_1 \\ v \end{bmatrix} \right] \\
& = (f_{0x}\tilde{p}_1 + H_{0xx}y_1 + H_{0xu}v_1)r_1 + (f_{0u}\tilde{p}_1 + H_{0ux}y_1 + H_{0uu}v_1)v
\end{aligned}$$

Now let's use the costate dynamics (42) of the auxiliary problem, together with its stationarity condition (41). Then we get:

$$\tilde{p}_1 \left[f_{0x}r_1 + f_{0u}v + [y_1, v_1] H_{0\sigma\sigma} \begin{bmatrix} r_1 \\ v \end{bmatrix} \right] = -\frac{dq_1}{dt}r_1 - q_1 [f_{0x}r_1 + f_{0u}v]$$

An integration by parts, together with the fact that $r_1(0) = q_1(T) = 0$ leads to:

$$R_3 = \alpha \int_0^T q_1 \left(\frac{dr_1}{dt} - [f_{0x}r_1 + f_{0u}v] \right)$$

But r_1 follows the dynamics (46), so that a Taylor expansion of $f(r_1 + x_0 + \alpha(\tilde{x}_1 + y_1), v + u_0 + \alpha v_1, t)$ at $(x_0 + \alpha(\tilde{x}_1 + y_1), u_0 + \alpha v_1)$ leads to:

$$\left(\frac{dr_1}{dt} - [f_{0x}r_1 + f_{0u}v] \right) = \int_0^1 \int_0^1 \lambda f_{\sigma\sigma}(x_0 + \alpha(\tilde{x}_1 + y_1) + \lambda\mu r_1, u_0 + \alpha v_1 + \lambda\mu v, t) \begin{bmatrix} r_1 \\ v \end{bmatrix}^2 d\lambda d\mu$$

so that, thanks to equation (33) about the bound of $f_{\sigma\sigma}$:

$$\left| q_1 \left(\frac{dr_1}{dt} - [f_{0x}r_1 + f_{0u}v] \right) \right| \leq \frac{1}{2}kM(|r_1| + |v|)^2 \leq kM(r_1^2 + v^2)$$

An integration on $[0, T]$ and a multiplication by α leads to

$$|R_3| \leq kM(T\|r_1\|_\infty^2 + \|v\|_2^2)$$

That equation, together with the bounds on r_1 (48) and v (45) lead to equation (117). $\square$

Proposition 34. *The following inequality holds:*

$$|R_4| \leq 2KTM(1 + k_{r5}\alpha)\alpha^2 \quad (118)$$

Proof. Let's develop $[y_1, v_1] H_{0\sigma\sigma} \begin{bmatrix} r_1 + \alpha\tilde{x}_1 \\ 0 \end{bmatrix}$ into coordinates:

$$[y_1, v_1] H_{0\sigma\sigma} \begin{bmatrix} r_1 + \alpha\tilde{x}_1 \\ 0 \end{bmatrix} = (H_{0xx}y_1 + H_{0uu}v_1)((r - r_1) + \alpha\tilde{x}_1)$$

Thus, thanks to the bound on $r - r_1$ (47), its absolute value can be bounded:

$$\left| [y_1, v_1] H_{0\sigma\sigma} \begin{bmatrix} r_1 + \alpha\tilde{x}_1 \\ 0 \end{bmatrix} \right| \leq 2KM(\|r - r_1\|_\infty + \alpha) \leq 2KM(k_{r5}\alpha^2 + \alpha) \leq 2KM(1 + k_{r5}\alpha)\alpha$$

An integration between 0 and T and a multiplication by α lead to equation (118). $\square$

J Proof of equation (55)

This is a consequence of equations (117) and (118), together with the fact that $R = R_3 + R_4$.

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